1

JIGS AND FIXTURES

1.1. GENERAL

Jigs and fixtures are special purpose tools which are used to facilitate production (machining, assembling and inspection operations) when workpieces are to be produced on a mass scale. The mass production of workpieces is based on the concept of interchangeability according to which every part will be produced within an established tolerance. Jigs and fixtures provide a means of manufacturing interchangeable parts since they establish a relation, with predetermined tolerances, between the work and the cutting tool. They eliminate the necessity of a special set up for each individual part. Once a jig or fixture is properly set up, any number of duplicate parts may be readily produced without additional set up. Hence jigs and fixtures are used :

1. To reduce the cost of production, as their use eliminates the laying out of work and setting up of tools.
2. To increase the production.
3. To assure high accuracy of the parts.
4. To provide for interchangeability.
5. To enable heavy and complex-shaped parts to be machined by being held rigidly to a machine.
6. Reduced quality control expenses.
7. Increased versatility of machine tool.
8. Less skilled labour.
9. Saving labour.
10. Their use partially automates the machine tool.
11. Their use improves the safety at work, thereby lowering the rate of accidents.

A jig may be defined as a device which holds and positions the work, locates or guides the cutting tool relative to the workpiece and usually is not fixed to the machine table. It is usually lighter in construction.

A fixture is a work holding device which only holds and positions the work, but does not in itself guide, locate or position the cutting tool. The setting of the tool is done by machine adjustment and a setting block or by using slip gauges. A fixture is bolted or clamped to the machine table. It is usually heavy in construction.

Jigs are used on drilling, reaming, tapping and counterboring operations, while fixtures are used in connection with turning, milling, grinding, shaping, planning and boring operations.

Jigs and fixtures, because of their functions and advantages are also called "Production Devices". To facilitate interchangeability, we need "Inspection Devices, *i.e.*, the different types of Gauges (Refer to Chapter 9 and 10)".

To fulfil their basic functions, both jigs and fixtures should possess the following components or elements:

1. A sufficiently rigid body (plate, box or frame structure) into which the workpieces are loaded.

2. Locating elements.
3. Clamping elements.
4. Tool guiding elements (for jigs) or tool setting elements (for fixtures).
5. Elements for positioning or fastening the jig or fixture on the machine on which it is used.

Locating pins are stops or pins which are inserted in the body of jig or fixture, against which the workpiece is pushed to establish the desired relationship between the workpiece and the jig or fixture. To assure interchangeability, the locating elements are made from hardened steel. The purpose of clamping elements is to exert a force to press a workpiece against the locating elements and hold it there in opposition to the action of the cutting forces. In the case of a jig, a hardened bushing is fastened on one or more sides of the jig, to guide the tool to its proper location in the work. However, in the case of a fixture, a target or set block is used to set the location of the tool with respect to the workpiece within the fixture. Most jigs use standard parts such as drill bushings, screws, jig bodies and many other parts. Fixutres are made from grey cast iron or steel by welding or bolting. Fixtures are usually massive bodies because they have to withstand large dynamic forces. Because the fixtures are in between the machine and the workpiece, their rigidity and the rigidity of their fastening to the machine table are most important. Jigs are positioned or supported on the machine table with the help of feet which slide or rest on the machine table. If the drill size is quite large, either stops are provided or the jig is clamped to the machine table to withstand the high drilling torque. Fixtures are clamped or bolted to the machine table. A simple jig and a fixture are shown in Fig. 1.1.

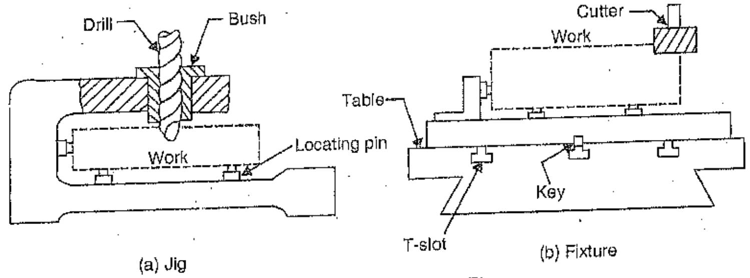


Fig. 1.1. A Simple Jig and a Fixture.

According to the degree of mechanization and automation, jigs and fixtures are classified as : (*a*) hand operated (*b*) power (*c*) semi-automatic (*d*) automatic.

1.2. LOCATING AND CLAMPING

The question of properly locating, supporting, and clamping the work is important since the overall accuracy is dependent primarily on the accuracy with which the workpiece is consistently located within the jig or fixture. There must be no movement of the work during machining. Locating refers to the establishment of a proper relationship between the workpiece and the jig or fixture. The function of clamping is to exert a force to press the workpiece against the locating surfaces and hold it there against the action of cutting forces.

1.2.1. Principle of Location. In order to study the complete location of a workpiece within a jig or fixture, let us consider a workpiece in space (Fig. 1.2). The workpiece is assumed to have true and flat faces. In a state of freedom, it may move in either of the two opposed directions along three mutually perpendicular axes, *XX*, *YY* and

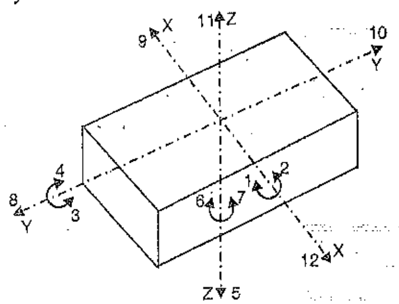


Fig. 1.2. Workpiece in Space.

ZZ. These six movements are called "movements of translation". Also, the workpiece can rotate in either of two opposed directions around each axis, clockwise and anticlockwise. These six movements are called "rotational movements". The sum of these two types of movements gives the twelve degrees of freedom of a workpiece in space. To confine the workpiece accurately and positively in another fixed body (jig or fixture), the movement of the workpiece in any of the twelve degrees of freedom must be restricted. For this, let us refer to Fig. 1.3.

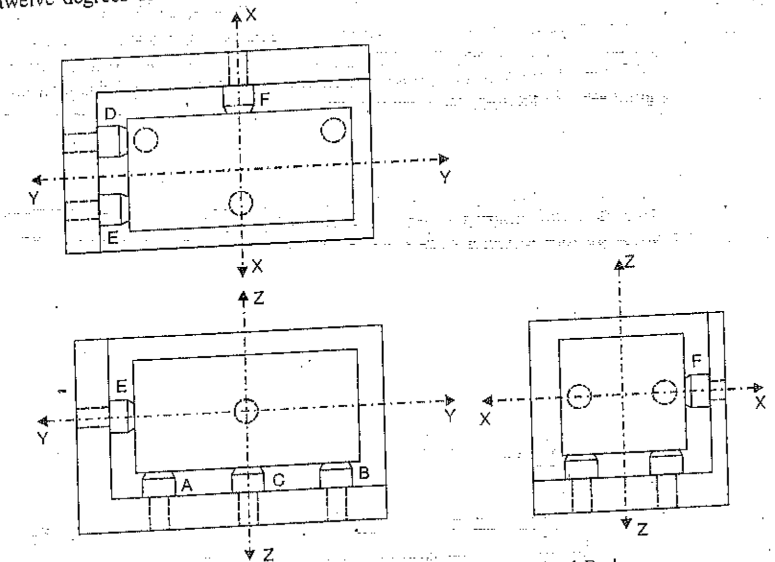


Fig. 1.3. Workpiece Located in a Fixed Body.

(*a*) The workpiece is resting on three pins *A*, *B* and *C* which are inserted in the base of the fixed body.

The workpiece cannot rotate about the axes *XX* and *YY* and also it cannot move downward. In this way, the five degrees of freedom 1, 2, 3, 4 and 5 have been arrested.

(*b*) Two more pins *D* and *E* are inserted in the fixed body, in a plane perpendicular to the plane containing the pins *A*, *B* and *C*. Now the workpiece cannot rotate about the *Z*-axis and also it cannot move towards the left. Hence, the addition of pins *D* and *E* restrict three more degrees of freedom, namely 6, 7 and 8.

(*c*) Another pin *F* in the second vertical face of the fixed body, arrests degree of freedom 9.

Thus, six locating pins, three in the base of the fixed body, two in a vertical plane and one in another vertical plane, the three planes being perpendicular to one another, restrict nine degrees of freedom. Three degrees of freedom, namely, 10, 11 and 12 are still free. To restrict these, three more pins, one for each of these degrees of freedom are needed. But this will completely enclose the workpiece making its loading and unloading into the jig or fixture impossible. Due to this, these remaining three degrees of freedom may be arrested by means of a clamping device. The above method of locating a workpiece in a jig or a fixture is called the "3-2-1" principle or "six point location" principle.

1.2.2. Principles of pin location. 1. **The principle of minimum locating points.** According to this principle, only the minimum locating points should be used to secure location of the workpiece in any one plane. Considering the "3-2-1" principle, there pins are used in the base of the fixed body.

This is due to the reason that this is the minimum number of locating points through which a plane can be drawn on which the workpiece will seat. The workpiece may rock and get strained if more than three locating points are provided. Now considering the second plane, it is clear that if one locating point is provided, the workpiece will swivel about this point, but not if there are two such location points. These two locating points establish a line parallel to the first plane. With the workpiece located against a plane and a line, it has only one direction of movement in third plane. Therefore, one locating point is sufficient in the third plane to prevent this movement.

2. **The principle of mutually perpendicular planes.** The "3-2-1" principle can also be put as : a workpiece may be fully located by supporting it against three points in one plane, two points in second plane and one point in a third plane. These three planes are not parallel and are preferably perpendicular to one another. If the locating surfaces are not perpendicular to one another, the following two difficulties will arise :

(*i*) The workpiece will tend to lift due to the wedging action between the two locating surfaces.

(*ii*) A large error in the movement of the workpiece introduced due to the displacement of a locating point or a particle (chip or dirt) adhering to it, as is clear from Fig. 1.4. The difference of resulting error and introduced error, that is, the projection factor is zero when the locating surfaces are normal and increases as the angle between them becomes more acute.

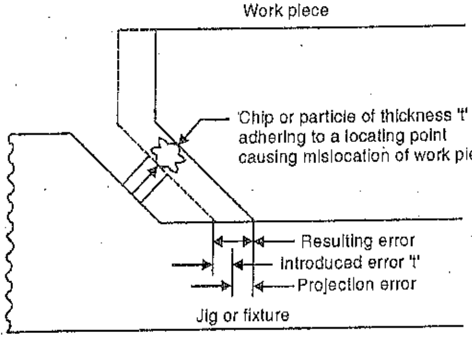


Fig. 1.4. Principle of Mutually Perpendicular Planes.

3. **The principle of extreme positions.** The locating points should be placed as far away from one another as possible, to achieve the greatest accuracy in location.

1.2.3. Locating devices. Pins of various designs and made of hardened steel are the most common locating devices used to locate a workpiece in a jig or fixture. The shank of the pin is press fitted or driven into the body of the jig or fixture. The locating diameter of the pin is made larger than the shank to prevent it from being forced into the jig or fixture body due to the weight of the workpiece or the cutting forces. Depending upon the mutual relation between the workpiece and pin, the pins may be classified as :

1. Locating pins 2. Support pins 3. Jack pins

1. **Locating pins.** When reamed or finelly finished holes are available in the workpiece, these can be used for locating purposes in the manner shown in Fig. 1.5. Depending upon their form, the locating pins are classified as :

(*i*) *Conical locating pins.* These pins are used to locate a workpiece which is cylindrical and with or without a hole as shown in Fig. 1.5 (*a*) and (*b*). Any variation in the hole size will be easily accommodated due to the conical shape of the pin.

(*ii*) *Cylindrical locating pins, 1.5* (*c*), (*d*) *and* (*e*) : In these pins, the locating diameter of the pin is made a push fit with the hole in the workpiece, with which it has to engage. The top portion of these pins is given a sufficient lead either by chamfering [Fig. 1.5 (*c*) and 1.5 (*d*)] or by means of radius [Fig. 1.5 (*e*)] to facilitate the loading of the workpiece.

2. **Support pins, Fig. 1.6.** With these pins (also known as rest pins, buttons or pads), workpieces with flat surfaces can be supported at convenient points. In the fixed type of support

pins, the locating surface is either ground flat [Fig. 1.6 (*a*)] or is curved [Fig. 1.6 (*b*)]. Support pins with flat head are usually employed to provide location and support to machined surfaces, because more contact area is available during location. It would ensure accurate and stable location and would not indent the work. The spherical head or rounded-head rest buttons are conventionally used for supporting rough surfaces (unmachined and cast surfaces), because they provide a point support which may be stable under these circumstances. Adjustable type support pins [Fig. 1.6 (*c*) and Fig. 1.6 (*d*)] are used for workpieces whose dimensions can vary, *e.g.*, sand castings, forging or unmachined faces.

If the component is to be located in the jig/fixture body, without the aid of these support pins, then the surface of the jig/fixture body where the component will be supported, will have to be machined. This will involve unnecessary machining time. The use of support pins saves machining time as only seats for the pins can be machined instead of the entire body of a large fixture. For small workpieces, however, no support pins are necessary. The fixture body itself can be machined suitably to provide the locating surfaces. An ample recess should be provided in corners so that burr on the workpiece corners or dirt and swarf do not obstruct proper location through positive contact of the workpiece with the locating surface. Support pins in large fixtures automatically provide similar recesses.

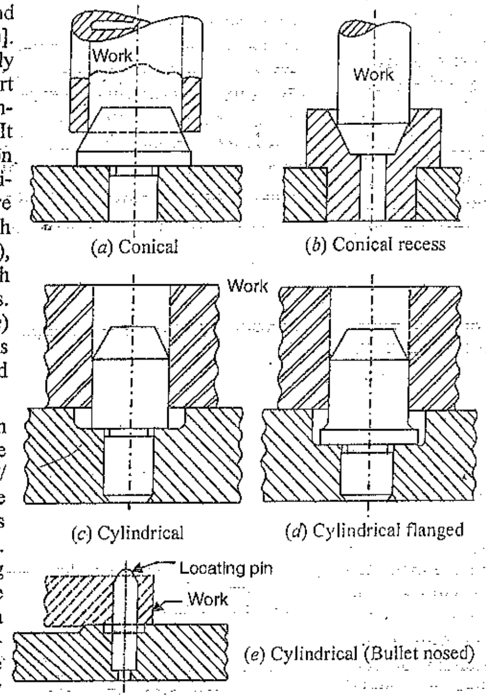


Fig. 1.5. Locating Pins.

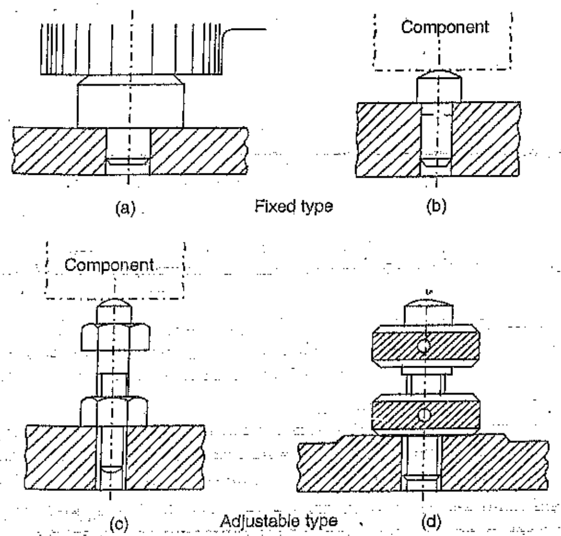


Fig. 1.6. Support Pins.

3. **Jack pins.** Jack pins or spring pins are also used to locate the workpieces whose dimensions are subject to variation, Fig. 1.7. The pin is allowed to come up under spring pressure or conversely is pressed down by the workpiece. When the location of the workpiece is secured, the pin is locked in this position by means of the locking screw.

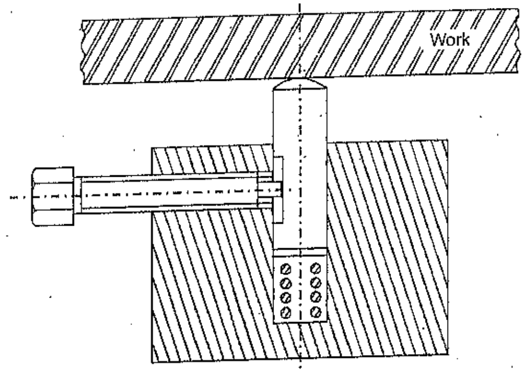


Fig. 1.7. Jack Pin.

1.2.4. Radial or Angular Location. Workpieces such as connecting rod or lever, which have two previously machined and finished holes at the two ends, may be located with the help of two pins projecting from the base surface of a jig or fixture, which will fit into the two holes in the workpiece, Fig. 1.8. Assuming that the workpiece is effectively located on pin A, the only movement the workpiece can have is that of rotation about the pin A. Now, neither the workpiece nor the jig or fixture can be made to the exact dimensions. It means the centre distances between pins A and B and between holes A and B are subject to variation. Let the tolerance for the centre distance between the holes A and B be 'x' and that for the centre distance between the pins, A and B by 'y'. Then if the workpiece is effectively located on pin A and if the pin B is a complete cylinder, the allowance between pin B and hole B will be x plus y. When the centre distance dimensions for the pins and holes are at maximum and minimum conditions, a large allowance will result between the hole and pin at B in the Y direction. Due to this, the workpiece will have undesirable rotation about the pin A and the pin B becomes useless. Therefore, to locate the component completely, location faces opposed to this rotational movement should be provided at the hole B. This is achieved by relieving the pin B

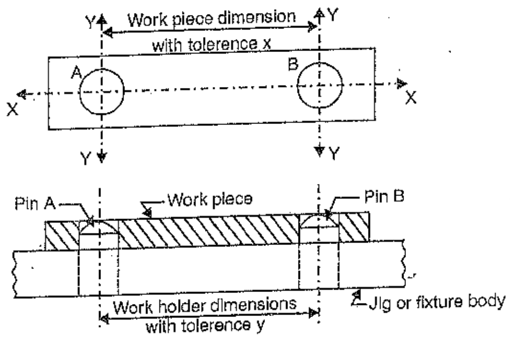


Fig. 1.8. Radial Location.

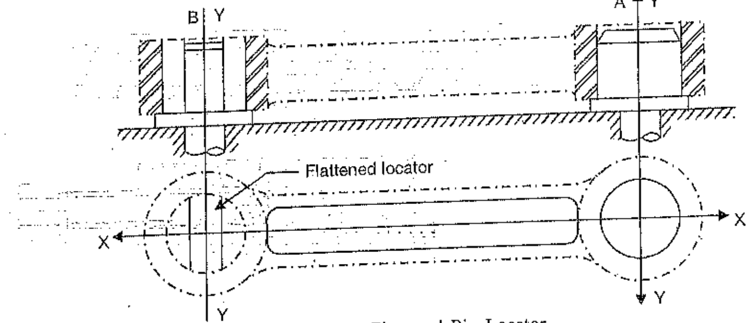


Fig. 1.9. Flattened Pin Locator.

on two sides perpendicular to the X-axis. This will allow for variations in the X-direction but will provide cylindrical locating surfaces in the Y-direction. This will result in a flattened or diamond pin locator as shown in Fig. 1.9 and Fig. 1.10 respectively.

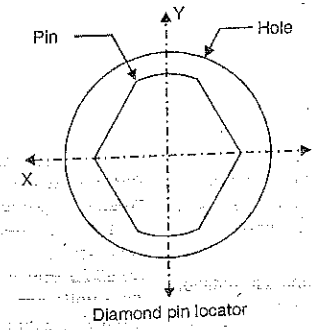


Fig. 1.10. Diamond Pin Locator.

The important and accurate hole of the two holes should be used for principal cylindrical location with a full cylindrical pin. The diamond pin is used to constrain the pivoting of the workpiece around the principal location. The principal locator should be longer than the diamond pin so that the workpiece can be located and pivoted around it before engaging with the diamond pin. This simplifies and speeds up locating of the workpiece.

A workpiece with only one hole can also be fully located as shown in Fig. 1.11. The principal location is secured from pin A. The radial movement in both the directions of Y-axis is restricted by providing two pins B confining the periphery of the workpiece. The basic principle for radial locations so as to minimize deviations from true locations is to position the radial locators as far from the axis of rotation as possible. This is clear in Fig. 1.12.

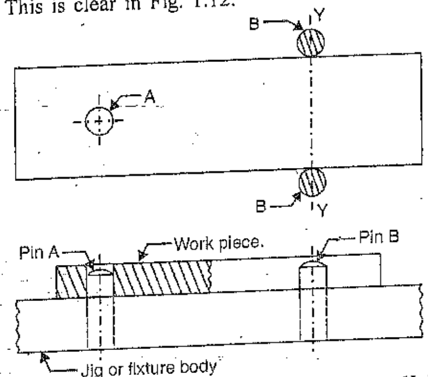


Fig. 1.11. Location of Workpiece with only one Hole.

A displacement 'd' at a distance 'a' from the axis O results in an angular error of AOA'. The same displacement 'd' at a greater distance 'b' gives an angular error of BOB' *which is smaller.*

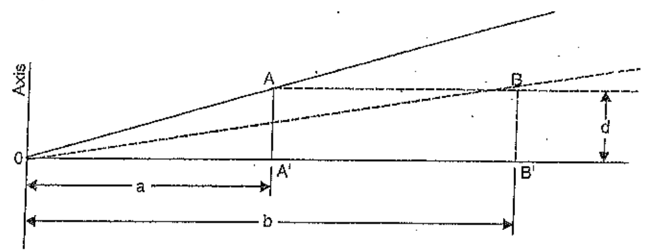


Fig. 1.12

1.2.5. *V*-location. In *V*-location, workpieces having circular or semicircular profile are located by means of a Vee block. The *V*-block should be used correctly so that the variations

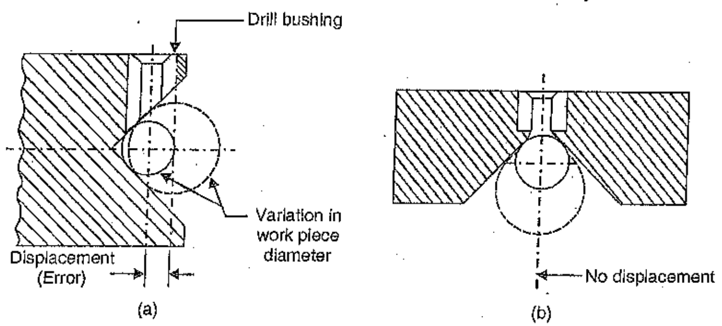


Fig. 1.13. *V*-location.

in the work piece size are not detrimental to location (Fig. 1.13). Vees can be used both for locating and clamping a workpiece. For this two Vees are employed, one fixed and the other sliding one. The fixed *V* acts to locate and the movable or sliding *V* acts to clamp and hold the workpiece at one end and forces it against the fixed *V* at the other end. To secure double clamping effect, the Vees may be made with inclined locating surfaces, instead of these being perpendicular to the direction of location of clamping. With inclined faces of the Vees a vertical downward component of the clamping force is obtained in addition to its horizontal component, Fig. 1.14. The vertical force component presses the workpiece on the base of the jig or fixture. The usual inclination of the face is 3°. The fixed *V* is secured to the jig or fixture body by means of caphead screws or dowel pins. The sliding *V* block may be actuated by means of a hand operated screw (Fig. 1.15) or a cam.

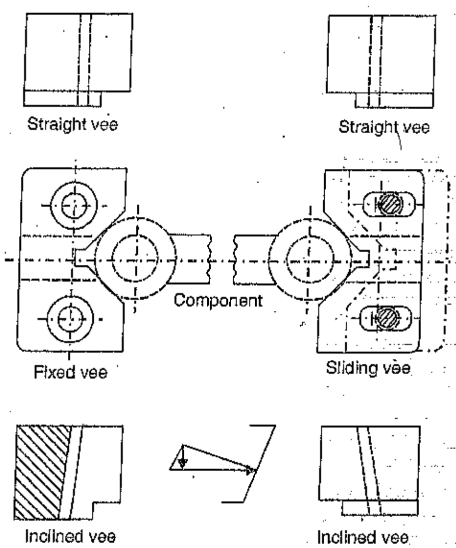


Fig. 1.14. Fixed and Sliding Vees

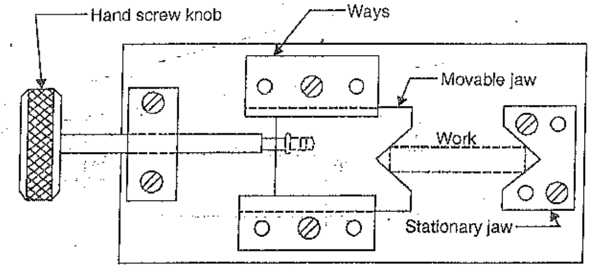


Fig. 1.15. *V*-block.

1.2.6. Bush location. Shaft type workpieces can be easily located in hardened steel bushes. Small and medium sized bushes are usually press fitted into the jig or fixture body whilst the large bushes are push fitted in the body and located by means of screws. The bushes can be plain or flanged type. A flange strengthens the bush and also prevents it from being driven into the jig body if it is left unlocked. In all the bushes, the entrance of the bush is chamfered, coned or bell mouthed to facilitate loading of the workpiece. A typical bush location is shown in Fig. 1.16.

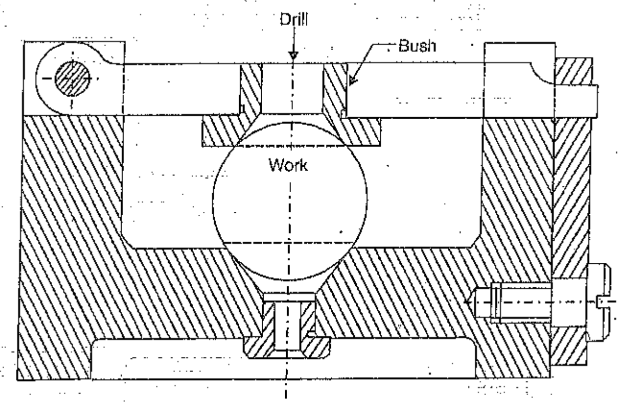


Fig. 1.16. Bush Location.

1.2.7. Design principles for location purposes. In addition to the principles discussed under Art. 1.2.2, the following principles should be followed while locating surfaces :

1. At least one datum or reference surface should be established at the first opportunity, from which subsequent machining will be measured.

2. For ease of cleaning, locating surfaces should be as small as possible consistent with adequate wearing qualities. Also, the location must be done from the machined surface.

3. The locating surfaces should not hold swarf and thereby misalign the workpiece. For this, proper relief should be provided where burr or swarf will get collected, as explained in Fig. 1.17.

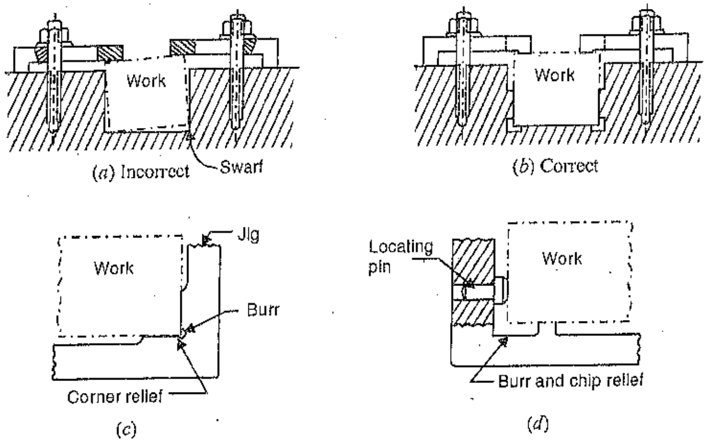


Fig. 1.17. Provision of Relief.

4. Locating surfaces should be raised above surrounding surfaces of the jig or fixture, so that chips fall or can be swept off readily, Fig. 1.18.

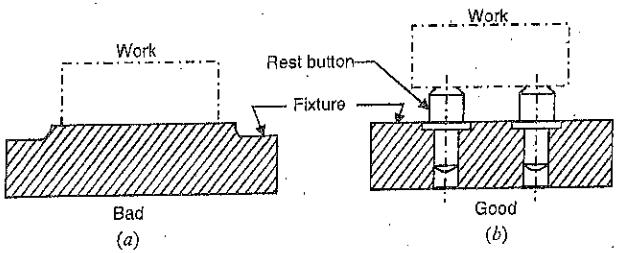


Fig. 1.18

5. Sharp corners in the locating surfaces must be avoided.

6. Adjustable type of locators should be used for the location on rough surfaces.

7. Locating pins should be easily accessible and visible to the operator.

8. To avoid distortion of the work, it should be supported as shown in Fig. 1.19.

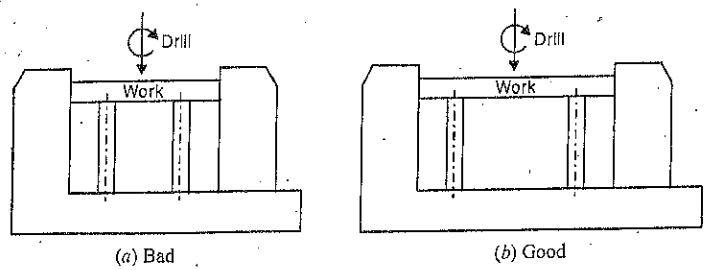


Fig. 1.19.

1.2.8. Clamping. If the workpiece cannot be restrained by the locating elements, it becomes necessary to clamp the workpiece in jig or fixture body. As already noted, the purpose of clamping is to exert a pressure to press a workpiece against the locating surfaces and hold it there in opposition to the cutting forces i.e to secure a reliable (positive) contact of the work with locating elements and prevent the work in the fixture from displacement and vibration in machining. The most common example of a clamp is the bench vise, where the movable jaw of the vise exerts force on the workpiece thereby holding it in the correct position of location in the fixed jaw of the vise.

Principles for clamping purposes. Since the proper and adequate clamping of a workpiece is very important, the following design and operational factors should be taken care of :

1. The clamping pressures applied against the workpiece must counteract the tool forces.

2. The clamping pressures should not be directed towards the cutting operation. Whenever possible, it should be directed parallel to it, Fig. 1-20.

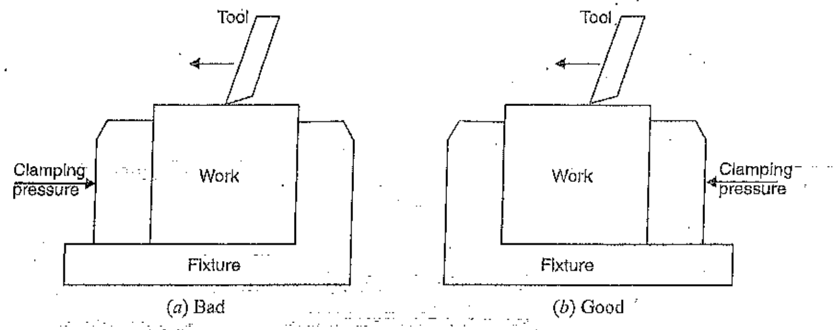


Fig. 1.20

3. The clamping pressure must only hold the workpiece and should never be great enough so as to damage, deform or change any dimensions of the workpiece.

4. The clamping and cutting forces should be directed towards the locating pins, otherwise the workpiece may get bent or forced away from the locating pins during machining.

5. Clamping should be simple, quick and foolproof. Complicated clamps lose their effectiveness as they wear.

6. The movement of a clamp should be strictly limited and if possible it should be positively guided.

7. Whenever possible, the lifting of the clamp by hand should be avoided if it can be done by means of a spring fitted to it.

8. Clamps should never be relied upon for holding the workpiece against the cutting force. The cutting force should be arranged against a fixed stop or a substantial part of the fixture body.

9. The clamps should always be arranged directly above the points supporting the work, otherwise the distortion of the work can occur, as illustrated in Fig. 1.21.

10. Fibre pads should be riveted to the clamp faces, otherwise soft and fragile workpiece can get damaged.

11. A clamp should be designed to deliver the required clamping force when operated by the smallest force expected.

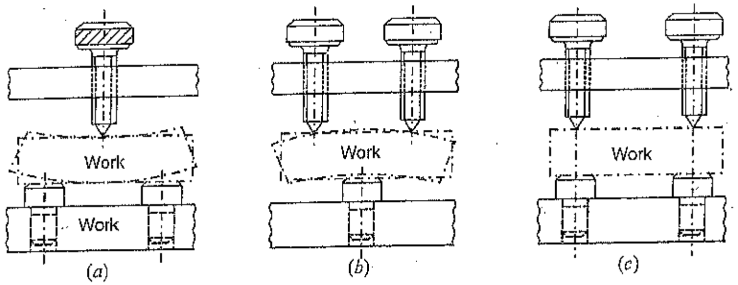


Fig. 1.21. Position of Clamp.

12. A clamp should be strong enough to withstand the reaction imposed upon it when the largest expected operating force is applied.

13. Clamping pressure should be directed towards the points of support, otherwise work will tend to rise from its support, Fig. 1.22.

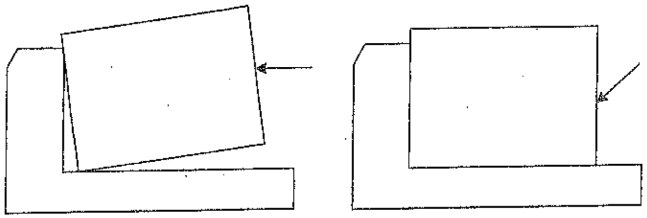


Fig. 1.22

1.2.9. Clamping Devices. The commonly used clamping devices are discussed below.

1. **Clamping screws.** Clamping screws are used for light clamping and typical examples are shown in Figs. 1.17 and 1.21.

2. **Hook bolt clamp.** This is very simple clamping device and is only suitable for light work and where the usual type of clamp is inconvenient. A typical hook bolt clamp is shown in Fig. 1.23.

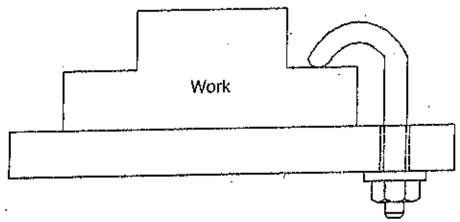


Fig. 1.23. Hook Bolt Clamp.

3. **Lever type clamps.** The various designs in the lever type clamp used with jigs and fixtures are discussed below.

(*i*) *Bridge clamp.* It is very simple and reliable clamping device. The clamping force is applied by the spring loaded nut Fig. 1.24 (*a*).

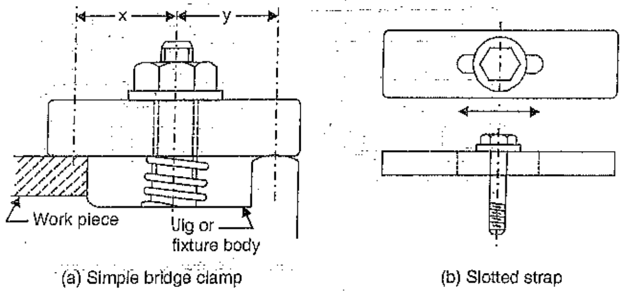


Fig. 1.24. (*a, b*)

The relative positions of the nut, the point of contact of the clamp with the work and with outer support should be carefully considered, since the compressive force of the nut is shared between the workpiece and the clamp support inversely as the ratio of their distances from the nut. The distance 'x' is less than or equal to but never greater than the distance 'y'. The spring is fitted with the clamp for its automatic lifting when the nut is loosened to remove the workpiece from the jig or fixture. To avoid the complete removal of the nut every time a workpiece is changed the clamp may be slotted to draw it back as shown in Fig. 1.24 (*b*). A two way clamping can be obtained by the bridge clamp as shown in Fig. 1.24 (*c*).

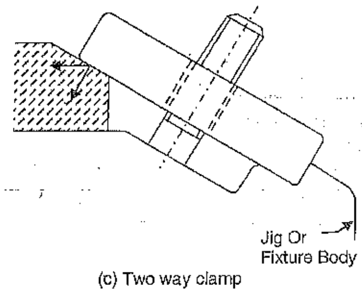


Fig. 1.24. Bridge Clamps.

(*ii*) *Heel clamps.* The various types of heel clamps are shown in Fig. 1.25. These consist of a robust plate or strap, centre stud and a heel. The strap should be strengthened at the point where the hole for the stud is cut out, by increasing the thickness around the hole. The design [Fig. 1.25 (*a*)] differs from the simple bridge clamp [Fig. 1.24 (*a*)] in that a heel is provided at the outer end of the clamp to guide its sliding motion for loading and unloading the workpiece. In design [Fig. 1.25 (*b*)], the heel is solid and one piece with the clamp. The workpiece is loaded into the jig or fixture or removed from these, by rotating the clamp. In design [Fig. 1.25 (*c*)], the clamp is guided by the loose heel which

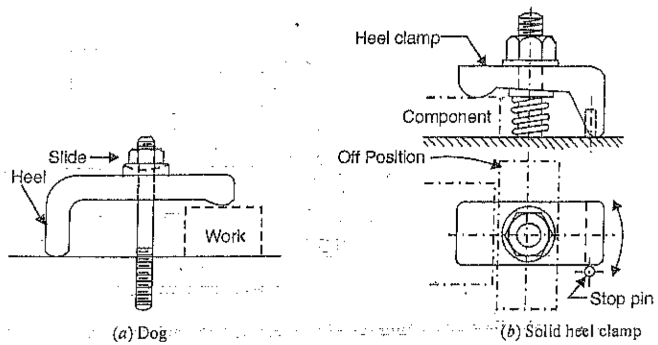
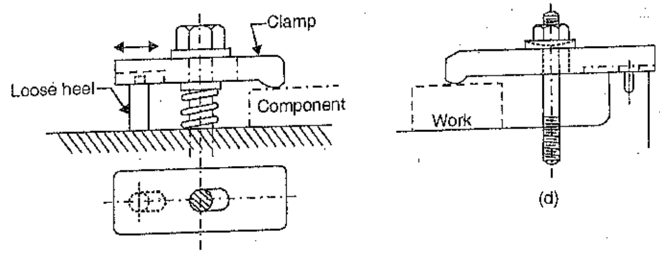


Fig. 1.25 (*a, b*)

is driven into the jig or fixture body. A short stem is turned on the end of the heel which fits loosely into a keyway in the clamp strap. The loading and unloading of the workpiece is obtained by reciprocating the clamp by hand. The design [Fig. 1.25 (*d*)] is similar to that in Fig. 1.25 (*c*) but, here the stem is provided at the end of the heel which forms part of the jig or fixture body casting.



(c) Loose guided heel clamp

Fig. 1.25. (*c, d*) Heel Clamps.

(*iii*) *Swinging strap* (*latch*) *clamp.* This is a special type of clamp which provides a means of entry for loading and unloading the workpiece. For this, the strap (latch or lid) can be swung out or in. Two designs of swinging latch clamps are shown in Fig. 1.26.

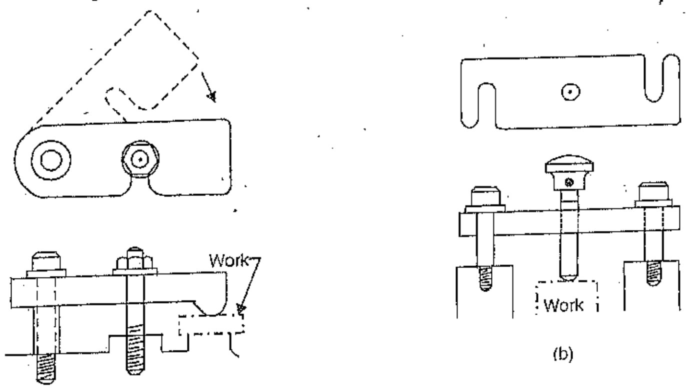
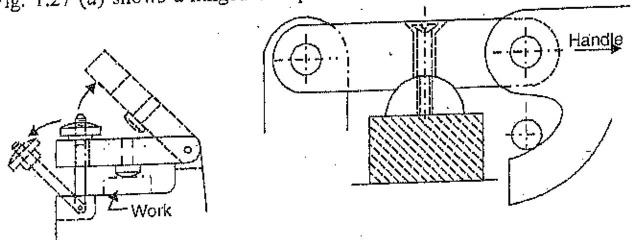


Fig. 1.26. Swinging Latch Clamp.

(*iv*) *Hinged clamps.* This clamp is similar to swinging latch clamp in which the latch is hinged to enable the workpiece to be loaded and unloaded. The clamp can be made integral with the latch. Fig. 1.27 (*a*) shows a hinged clamp which is locked by means of a bolt. Fig. 1.27 (*b*)



(*a*) Hinged Clamp (*b*) Cam-operated Clamp

Fig. 1.27. Hinged Clamps.

shows a hinged clamp provided with a hook cam. This clamp is much quicker than the bolt type and is suitable for workpieces which maintain dimensional accuracy. The hooked end of the operating lever acts as a cam and engages a pin. Fig. 1.28 shows some other designs of the lids or straps which may be used for swinging latch or hinged clamps.

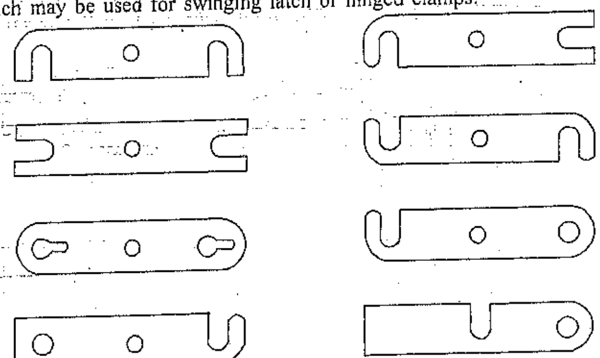
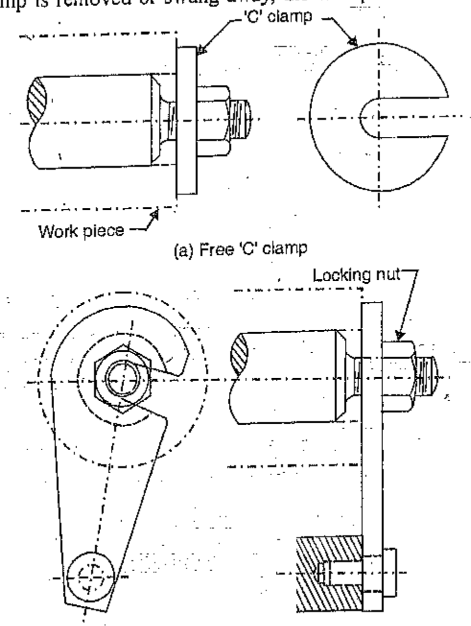


Fig. 1.28. Straps.

4. **Quick acting clamps.** There are many mechanical clamping devices (pneumatic and hydraulic devices will be discussed later), which can be termed as quick acting. These devices are costlier than the other types but ultimately prove economical since these help in reducing the total operating time. Some of the quick acting clamping devices are discussed below :

(*i*) *C-clamps.* The two types of C-clamps, free and captive are shown in Fig. 1.29. To unload the workpiece, the locking nut is unscrewed by giving it about one turn and this releases the C-clamp. When the clamp is removed or swung away, the workpiece can freely pass over the nut.



(b) Captive 'C' clamp

Fig 1.29. *C* - clamps.

The reverse procedure is adopted for loading the workpiece. The free C-clamp may be fastened to jig or fixture body to prevent it from being lost.

(*ii*) *Quick acting nut.* This nut is shown in Fig. 1.30. The threads of the nut are not continuous but are interrupted. The length of the nut is about 2 to 3 times the thread diameter. The diameter of the clearance '*D*' is slightly bigger than the outside diameter of the thread and the axis of the hole is inclined at angle (3° to 7°) to the axis of nut. The use of the quick acting nut is explained in Fig. 1.31. When the nut is assembled over the male thread, it is inclined to the axis of the clearance hole. When the nut engages the male thread, it is dropped on to the screw threads and is then tightly locked by giving it about half a turn.

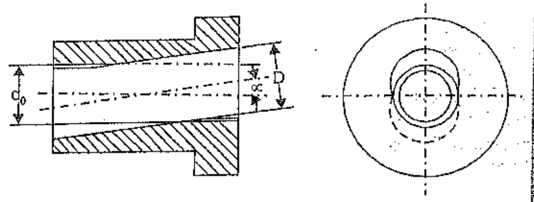


Fig. 1.30. Quick Acting Nut.

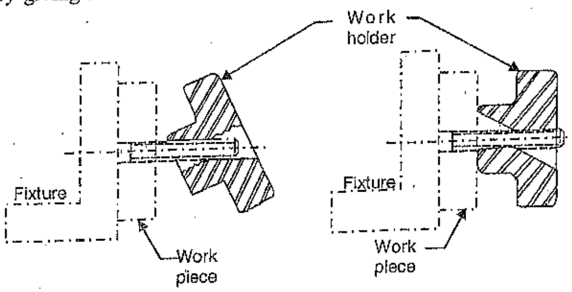


Fig. 1.31. Use of Quick Acting Nut.

(*iii*) *Cam-operated clamp.* These clamps find broad application and are fast and positive in action. These should not be used where vibrations are present or where the dimensions of the workpiece vary, *e.g.*, sand castings. A cam operated clamp is shown in Fig. 1.32.

Note. No clamping devices are used if a very heavy stable job is to be machined, whose weight is very great compared to the forces developed in the cutting process, if these forces are in a direction that cannot disturb the setting of the job (as, for example, in drilling holes in a heavy baseplate). Clamping devices are also unnecessary if the job is deprived of all of its degrees of freedom when it is loaded into a fixture (as, for example, turning a job between centres on a centre lathe and milling the two end faces of a connecting rod located on two pins from its two end holes, Fig. 1.9).

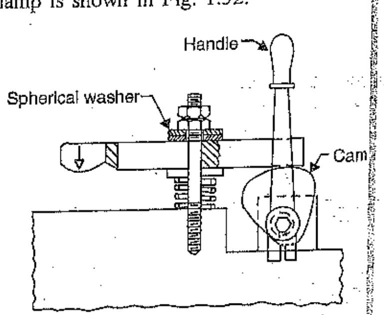


Fig. 1.32. Cam-operated Clamp

1.2.10. Materials for Locating and Clamping Elements. The locating and clamping elements are generally made from steel. The elements which come in contact with the workpieces or are subjected to wear, should be hardened and wherever necessary, the working face should be ground. For dowel pins and handles, silver steel is generally used. For complicated shapes and when exceptional wear is liable to occur, good quality case-hardened steel, tool steel or a high tensile steel may be used.

1.2.11. Lever type clamps and spherical washers. In the lever type clamps discussed above, it is seen that the clamping face of the lever is curved. This makes the clamp operatable

even if there is variation in the workpiece. At the other end of the strap (pressure pad), the top of the bridge or the heel should also be in the shape of raised and rounded toes to permit some tilting of clamp. This design provides more effective clamping than a design having flat strap ends. This design will also take care of small variations in workpiece height.

Poor clamping conditions can result if there is a considerable variation in the workpiece or there is difference in workpiece and fulcrum block height. The misalignment between clamp surface and clamping nut due to tilting of clamp can be taken care of by interposing a pair (male-female) of spherical washers between the nut and the strap (instead of a plain washer), Fig. 1.33. The spherical bearing surfaces of the washers will allow the inclination of the strap caused by the difference in heights of the filcrum block and workpiece. The male washer (upper one) remains square with the nut while the female washer (lower one) tilts with the clamp; since the spherical bearing surfaces allow the pair of spherical washers to tilt with respect to each other. The angle of inclination of the strap that can be tolerated is limited by the clearance between the stud and the inside diameter of the washers. Spherical washers are thus commonly used for equalising clamping forces.

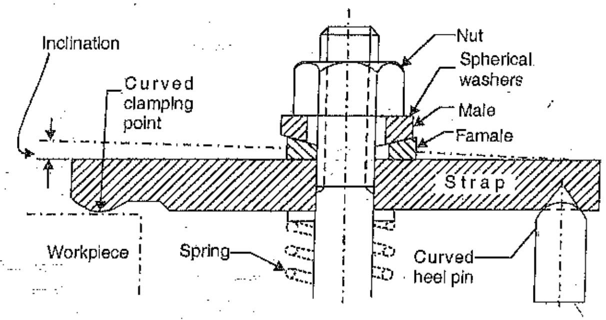


Fig. 1.33. Use of Spherical Washers.

1.3. DESIGN PRINCIPLES COMMON TO JIGS AND FIXTURES

1. Since the total machining time for a workpiece includes work-handling time, the methods of location and clamping should be such that the idle time is minimum. The various design principles regarding location and clamping have already been discussed.

2. The design of jig and fixture should allow easy and quick loading and unloading of the workpiece. This will also help in reducing the idle time to minimise.

3. The jig and fixture should be as open as possible to minimize chip or burr accumulation and to enable the operator to remove the chips easily with a brush or an air jet.

4. *Fool proofing.* It can be defined as the incorporation of design features in the jig or fixture, that will make it impossible to load the work into the jig or fixture in an improper position but will not interfere with proper loading and locating the workpiece. There are many foolproofing devices such as fouling pegs, blocks or pins which clear correctly positioned parts but prevent incorrectly loaded parts from entering the jig or fixture body.

Figure 1.34 explains this principle. Three holes are to be drilled in the component shown. The operator locates the component on the bottom plate of the drilling jig with the projection *C* on the component fitting the locating hole in the jig. Now if the component is being located incorrectly so that the end *A* of the component is oriented towards the left of the locating hole, the fouling peg in the body of the jig will obstruct the component.

For correct locations, the end A of the component is to be towards the right and the curved end B of the component is to be towards the left of the locating hole.

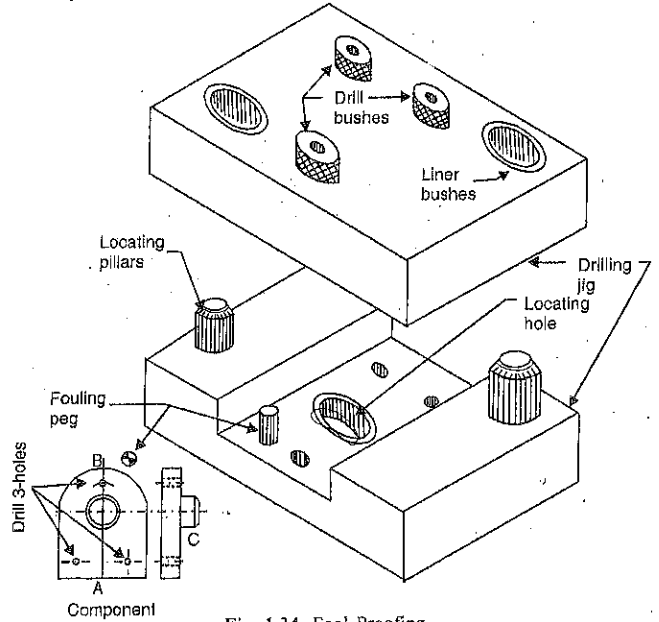


Fig. 1.34. Fool Proofing.

5. *Clearance.* Clearance is provided in the jig or fixture body for two main reasons :

(*i*) to allow for any variation in component sizes, especially castings and forgings.

(*ii*) to allow for hand movements so that the workpiece can easily be placed in the jig or fixture and removed after machining.

6. *Rigidity.* Jigs and fixtures should be sufficiently stiff to secure the preset accuracy of machining.

7. *Trunnions.* To simplify the handling of heavy jigs or fixtures, the following means can be adopted :

(*i*) Eye-bolts, rings or lifting lugs can be provided for the lifting of the jig or fixture.

(*ii*) If the workpiece is also heavy, then the jig design should allow for side loading and unloading by sliding the workpiece on the machine table.

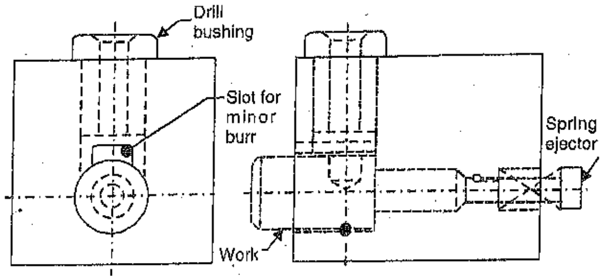


Fig. 1.35. Burr Grooves.

8. *Burr grooves.* A burr raised on the work at the start of a cut is termed a 'minor burr' and that at the end of a cut a 'major burr'. Jigs should be designed so that the removal of the workpiece is not obstructed by these burrs. For this, suitable clearance grooves or slots should be provided as shown in Fig. 1.35.

9. *Ejectors.* The use of ejection devices to force the workpiece out from the jig or fixture is important in two situations :

(*i*) the workpiece is heavy.

(*ii*) machining pressure forces the workpiece to the sides or base of the jig or fixture and the pressure and oil or coolant film will cause the work to stick and be difficult to remove.

On small jigs or fixtures, a pin located under the work will remove the part readily [Fig. 1.36 (*a*), (*b*)]. Hinged ejectors [Fig. 1.36 (*c*)] are also very useful and can be easily mounted.

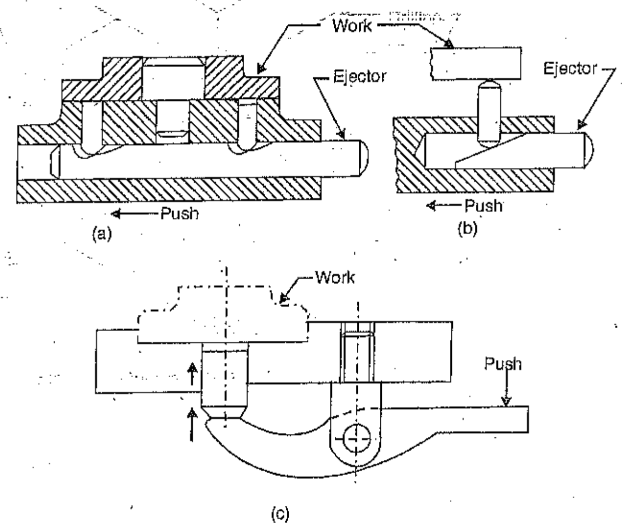


Fig. 1.36. Ejecting Devices.

10. **Inserts.** To avoid any damage to fragile and soft workpieces and also to the finished surfaces of a workpiece while clamping, inserts of some soft material such as copper, lead, fibre, leather, hard rubber, plastic or felt should be fitted to the faces of the clamps.

11. **Design for safety.** Jigs/fixtures must be safe and convenient in use. Following are some of the factors for the safety of the worker working with a jig/fixture :

(*i*) Sharp corners on the body of the jig/fixture should be avoided.

(*ii*) Sighting surfaces should be clear.

(*iii*) Bolts and nuts should be inside the body of the jig/fixture and not protrude on the surface.

12. **Sighting Surfaces.** Machining on a workpiece must be clearly visible to the worker. He should not be required to bend his neck for seeing the work surface.

13. **Simplicity in Design.** Design of the jig/fixtures should be a simple one. A complicated design requires a large maintenance. They should be cheap in manufacture and should lend themselves readily to maintenance and replacement of worn-out parts.

14. *Economical.* Jig/fixture should be simple in construction, give high accuracy, be sufficiently rigid and light in weight. To satisfy all these conditions, an economical balance has to be made.

15. They should be easy to *set in the* machine tool, which is so important in quatity production where jigs/fixtures are replaced at intervals.

1.4. DRILLING JIGS

Drilling jigs are used to machine holes in mechanical products. To obtain positional accuracy of the holes, hardened drill bushes or jig bushes are used to locate and guide drills, reamers etc., in relation to the workpiece. These guide bushes are not essential but these prove to be economical and technically desirable as will be discussed ahead. The portion of the jig into which the hardened bushes are fitted is called bush plate.

Drilling jigs are either clamped to the workpiece in which holes are to be drilled or the workpiece is housed and clamped in the jig body. If more than one hole is to be drilled, the drill jig is made to slide on the table of the drilling machine. Such a drill jig is moved by hand into position under the drill so that the drill readily enters the bush. During the drilling operation, the jig is held by hand. If the drill size is large enough to produce a high torque, either stops should be provided or the drill jig clamped to the table of the drilling machine. A drill jig is provided with feet which rest or slide on the table of drilling machine. These feet should be outside the cutting forces, thus providing solid support.

Drilling jigs make feasible the drilling of holes at higher speed, with greater accuracy and with less skilled workers than is possible when the holes are laid out and drilled "by hand". Also, they produce interchangeable parts, because each part drilled in a drilling jig should have the same hole pattern as every other part.

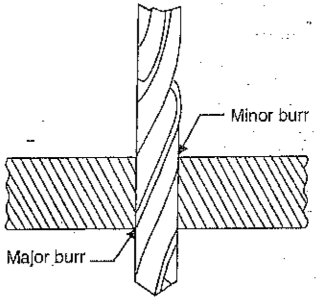


Fig. 1.37. Major and Minor Burrs.

It is clear that during the drilling operation, burrs will be produced. The burr produced at the start of a hole is smaller than that produced at the end of the hole. The first type is called 'minor burr' and the second type 'Major burr' (when the drill breaks through the material, Fig. 1.37). When designing a drilling jig, these two types of burrs should be taken into consideration since they may cause difficulty in unloading the workpiece from the jig after a hole has been drilled.

1.4.1. Design Principles for Drilling Jigs. 1. A drilling jig should be of light construction consistent with adequate rigidity to facilitate its handling because it has to be handled frequently during the operation.

2. A drilling jig which is not normally clamped to the machine table should be provided with four feet so that it will rock if it is not resting square on the machine table and so warn the operator.

3. The stability of a drilling jig should be as good as possible since it is not usual to clamp it to the machine table and to ensure this, the feet or base of the jig should extend well outside the holes to be drilled.

4. Drill bushings should be fitted in fixed portion of the jig and not in clamps except for a few special cases (for example, leaf type jig).

1.4.2. Drill Bushes. Sometimes the stiffness of the cutting tool may be insufficient to perform certain machining operations. To eliminate the elastic spring back in machining and to locate the tool relative to the work, use is made of guiding parts, such as, jig bushings and templates. These must be precise, wear resistant and changeable.

Jig bushings are used in drilling and boring jigs. Their use permits giving up the marking out, reduces drill run-off and hole expansion (ovalization). The diametric accuracy of holes in jig drilling is 50 per cent higher on the average compared to that of holes drilled conventionally. Drill bushings can be classified as : Press fit, Renewable and Liner bushings.

(*i*) *Press Fit bushings.* These bushings are used when little importance is put on accuracy or finish, and the tool used is a twist drill. These bushings are installed directly in the jig body and are used mainly for short production runs not requiring bush replacement. These are also employed where the centre distances of holes are too close to permit the fitting of liners and renewable bushes. There are two designs of press fit bushings : (*a*) Plain or headless (*b*) Flanged or headed. A flanged or headed bush has a flange or head, Fig. 1.38. (*b*). It is employed when the jig plate into which it is installed is thin, the flanged or headed portion increasing the length of the bush which provides longer guiding portion to the bush than would otherwise be available. The flange or head also acts as a stop for the tool.

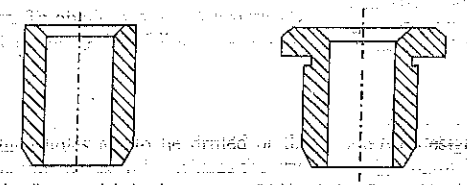


Fig. 1.38. Press Fit Bushings.

(*ii*) *Renewable bushes.* Figure 1.39 (*a*). When the guide bushes require periodic replacement (due to the wear of the inside diameter of the bush, in the case of continuous or large batch production), the replacement is simplified by using a renewable bush. These are of the flanged type and are sliding fit into the liner bush, which is installed (press fitted in the jig plate). The liner bush provides hardened wear resistant mating surface to the renewable bush.

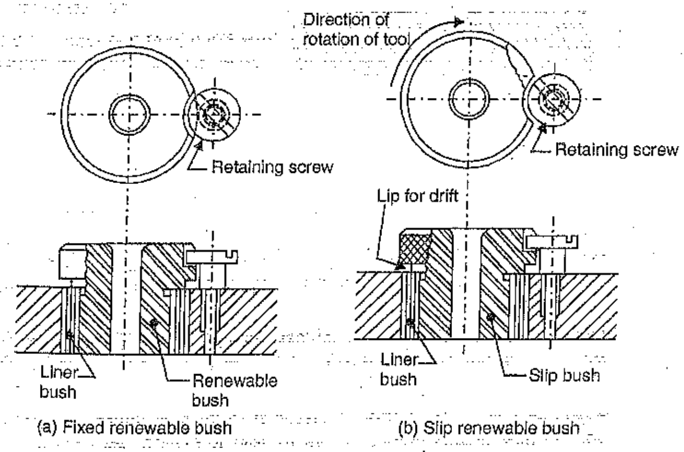


Fig. 1.39. Renewable Bushes.

The renewable bushes must be prevented from rotating or lifting with the drill. One common method is to use a retaining screw as shown in the Fig. 1.39 (*a*). The flange of the bush is provided with a flat. When the renewable bush is put into the liner bush and the retaining screw is tightened, the collar of the screw will press against the flat on the bush flange and prevent rotation of the bush and also its lifting up (when the drill is clearing the swarf or is being withdrawn at the end of the cut). When the renewable bush wears out, the retaining screw is removed and the worn bush is taken out. A new bush is then easily substituted. A normal press fit bush can be taken out only by driving it out of the jig plate. This usually damages the bore of the jig plate which has to be rebored for an oversize bush to be fitted.

(*iii*) *Slip Bushes.* Fig. 1.39 (*b*). Slip bushes are used when more than one bushings are to be interchanged in a given size of the liner, that is, where two or more operations in a job require different inside diameters of the guide bush in the same jig, for example, when drilling is following by reaming, counterboring etc. The hole is first drilled using a guide bush of the requisite inside diameter. After drilling, this bush is removed and another bush to guide the reamer is put within the liner bush. In mass production, these bushings should be changed with minimum loss of time. At the same time, the slip bushes should be prevented from rotating or getting lifted up during the machining process. Both these objectives are accomplished as follows :

A retaining screw is used which is fixed permanently in the bush plate. The slip bush is provided with a clearance slot with a radius slightly larger than that of the head of the screw. For loading and unloading of the slip bush, this slot is aligned with the collar of the retaining screw. The bush can be moved freely axially in this position. For loading, the bush can, therefore, be dropped over the screw. The bush is also provided with a step, which when the bush is rotated clockwise, will turn and lock under the flange of the screw. When the tool is withdrawn, the screw head prevents the bush from rising. For unloading, the bush is rotated anticlockwise to align the bush clearance slot with screw collar. Then the bush can be lifted up axially out of the liner. The slip bushings are also flange type and are sliding fit into the liner bush and for their easy loading/unloading, their heads are knurled.

(*iv*) *Screw bush.* The screwing of the bush into the jig body not only holds the bush in place, but it also makes the bush adjustable. This drill bush may also be used for locating purposes and is then invariably screwed into position; it can, therefore, be adjusted for length to suit the component. It can also be tightened down to give clamping pressure when required. The screw threads are not depended upon for accurate location. So, if an accurately positioned hole is required, it will be necessary to locate the bush in the liner bush on two spigots. This will ensure that the thread is used only for moving and not for positioning the drill bush. Alternately, guiding portion is provided on the bush body, Fig. 1.40 (*c*). The straight cylindrical guiding portion, is fitted into the hole in the drill plate, which is counterbored and tapped. The cylindrical portion may be placed above the screw threads if desired, in which case it will be of larger diameter than the threads. In either case, it must be held concentric with the hole within close enough limits to provide the required accuracy. If the position of the hole to be drilled is unimportant, the above refinements will be unnecessary. The screwed bush prevents its rotating and lifting up. A peg, Fig. 1.40 (*a*) can also achieve the same functions.

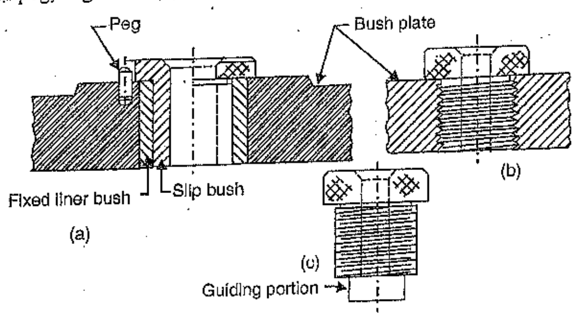


Fig. 1.40

(*v*) *Liner Bushings.* These bushings, also known as 'master bushings' are permanently fixed into the jig body. These act as guides for renewable type bushings. These bushings can be

with or without heads, Fig. 1.41. A liner bush is always used in conjunction with a renewable bush, for example a slip bush. Slip bushes of different inside diameters but of the same outside diameters (same as the inner diameter of the liner bush) are used.

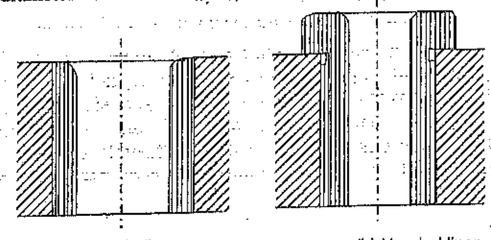


Fig. 1.41. Liner Bushings.

(*vi*) *Special drill bushings.* Some examples of special drill bushings are given in Fig. 1.42 :

(*a*) is used to drill a hole through an inclined surface.

(*b*) A long busing is used if the hole being drilled is in a recess. Since the bush is long, therefore, the drill friction is considerable. To reduce it, the bush can be counterbored (*i*). The other method is shown in (*ii*). The larger bush can be made of C.I. or a cheaper material, only the actual drill bush being of the normal material.

(*c*) If two holes are to be drilled close together, two bushings with flats can be used, or

(*d*) a bushing with two holes is used.

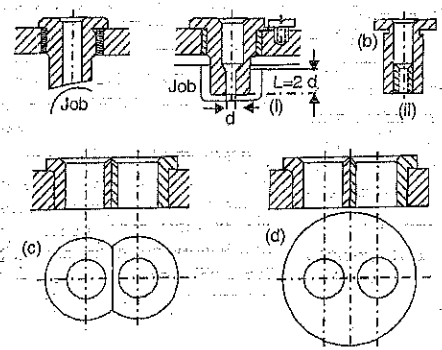


Fig. 1.42. Special Drill Bushings.

1.4.3. Design Principles for Drill Bushings. The following points should be kept in mind in the case of drill bushings :

1. To facilitate easy entry of drills, the entrances to drill bushes should be extremely smooth and well chamfered or rounded. Also, a suitable lead or chamfer should be on the outside of the bush at the lower end to facilitate its installation in the jig body (Fig. 1.38).

2. There should not be any sharp corners on the body of the bush.

3. Loose or screwed in solid bushes should not be used where accuracy is important.

4. The effective length of the drill bushing should be sufficient to guide and support the drill. A too short a bush will not be able to keep the drill in line as the drill can bend with the short bushing acting as a fulcrum. Due to this, out of line and oversize holes may be produced. On the other hand, if the bush is longer than necessary, the drill will wear out earlier. For drills having average helix angles, the length of the drill bushing should be from 1.75 to 2.50 times the drill diameter.

5. Adequate provisions must be made for the chips that are produced and for their easy removal. Insufficient clearance between the end of a drill bushing and the workpiece, Fig. 1.43 (*c*), will prevent the chips from escaping and these will come up through the drill bushing. This will result in the wear of the bushing due to the abrasive action of the chips. On the other hand, too much clearance, Fig. 1.43 (*a*), may not provide adequate drill guidance and can result in broken drills, due to bending between the bushing and the workpiece. Fig. 1.43 (*b*), shows the

correct design. The clearance between the lower end of the bushing and the workpiece should be from 1/3 to 1 times (lower values for C.I. and higher values for steel) and for drilling deep holes in steel, as much as 1.5 times the diameter of the hole being machined. This design reduces wear because the chips do not pass through the bushing but go-off sideways.

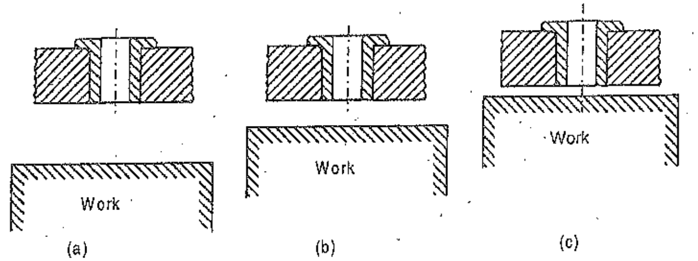


Fig. 1.43

6. The hole of the drill bushing should be from 0.00025 to 0.0025 cm larger than the drill size.

Note. Jig bushings can be used to drill from 10,000 to 15,000 holes.

1.4.4. Drill Bush Materials and Manufacture. The surface of the guide hole of the drill bush has to resist wear against the abrasive action of the chips and also of the drill. So, the drill bushes should be made of proper materials and good care should be taken for their manufacturing. Drill bushes are made either from good quality case hardening steel or of tool steel. They can also be made of mild steel which is carburised to give a sufficient case depth. After hardening, the drill bushes are ground, the bore and the outside diameter being ground concentric. The bore of the bush is sometimes lapped to give good finish and a fine running fit with the tool.

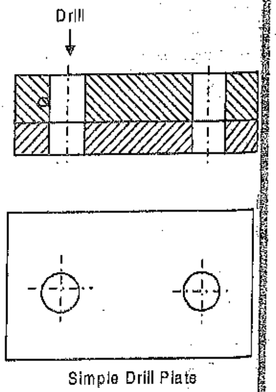


Fig. 1.44. Template Jig.

1.4.5. Types of drilling jigs. There are no hard and fast criteria for classifying the drilling jigs. However, the drilling jigs may be classified as follows :

1. *Template jig.* This is the simplest type of drilling jig. It is simply a plate made to the shape and size of the workpiece with the required number of holes made in it accurately. A simple template type of jig is shown in Fig. 1.44. It is placed on the workpiece and the holes in the workpiece will be made by the drill which will be guided through the holes in the template. The plate should be hardened to avoid its frequent replacement. This type of jig is suitable if only a few parts are to be made.

2. *Plate type jig.* This is an improvement over the template type of jig. In place of simple holes, drill bushes are provided in the plate to guide the drill. The workpiece can be clamped to the plate and the holes drilled, Fig. 1.45.

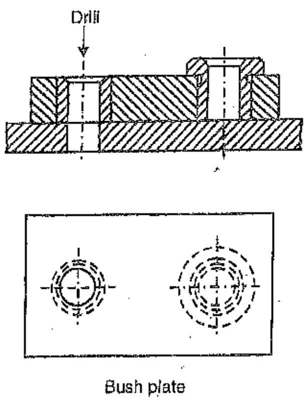


Fig. 1.45. Plate Type Jig

3. *Open type jig.* In this jig, Fig. 1.46, the top of the jig is open. The workpiece is placed on top.

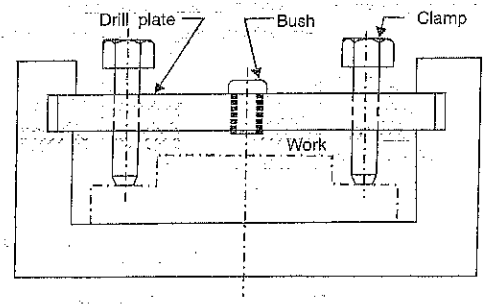


Fig. 1.46. Open Type Jig.

4. *Swinging leaf jig.* It is also a sort of open type jig in which the top plate is arranged to swing about a fulcrum point so that it completely clears the jig for easy loading and unloading of the workpiece, Fig. 1.47. The drill bushes are fitted into the plate which is also known as leaf, latch or lid.

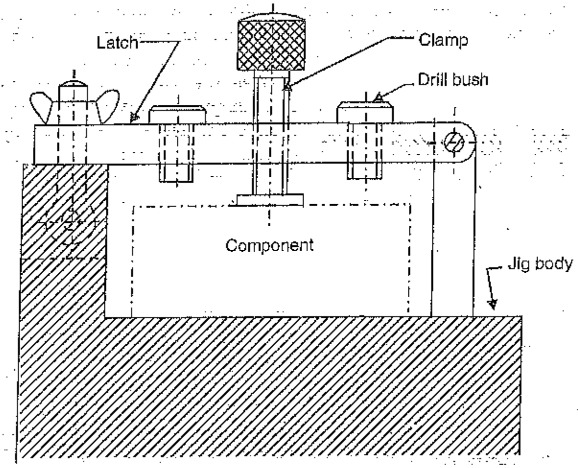


Fig. 1.47. Swinging Leaf Jig

5. *Box type jig.* When holes are to be drilled in more than one plane of a workpiece, the jig has to be provided with equivalent number of bush plates. For positioning the jig on the machine table, feet have to be provided opposite each drilling bush plate. One side of the jig will be provided with a swinging leaf for loading and unloading the workpiece. Such a jig would take the form of a box. (Fig. 1.48). The body of such a jig should be as light as possible since it will have to be lifted again and again. Fig. 1.48 is for a leaf-type box jig. When, one or more sides of the box jig are kept open for loading/unloading, it is known as Tumble type and Trunnion type box jig.

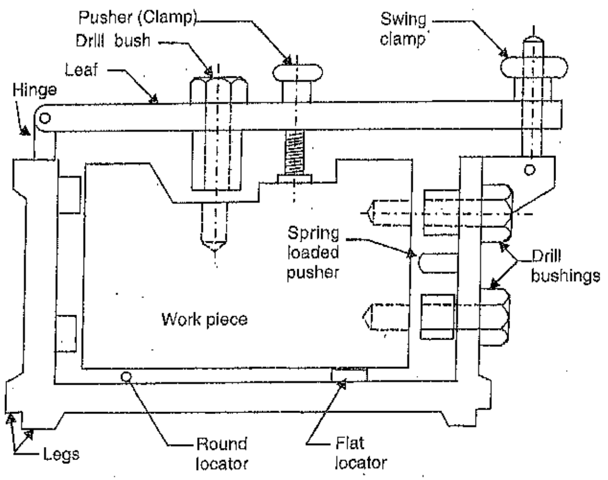


Fig. 1.48. Box Type Jig.

6. *Channel type jig.* In this type, the jig is made of standard steel channel section. This jig can also be provided with a swinging leaf to form a channel-and-leaf Jig (Fig. 1.49).

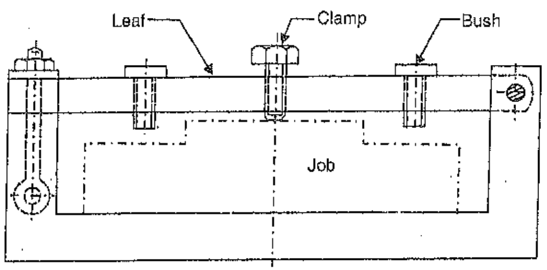


Fig. 1.49. Channel Type Jig.

7. *Sandwich jig:* A sandwich jig is a modification of a plate type of jig. The plate jig has a back up plate. The job is held between the two plates. The jig is very useful for thin and ductile jobs which might get bent or warped on another type of jig.

8. *Angular jig:* This type of jig is used when a hole is to be drilled at an angle to the drilling bush axis. Fig. 1.50. This type of jig is used to drill holes in collars and hubs of pulleys and gears etc. Fig. 1.50 refers to a drilling jig for drilling oil holes in an I.C. engine connecting rod.

9. *Angle plate jig:* This type of jig is used to drill holes in parts at right angles to their mounting locations. Fig. 1.56 can be an example of angle plate jig.

10. *Pot jig:* This type of jig is used for drilling holes in circular components, which have both internal and external diameters. The body of the jig is in the form of a pot. The workpiece is located in the pot of the jig and is properly clamped with the help of a post type locating pin, a clamping plate and a clamping device, Fig. 1.51.

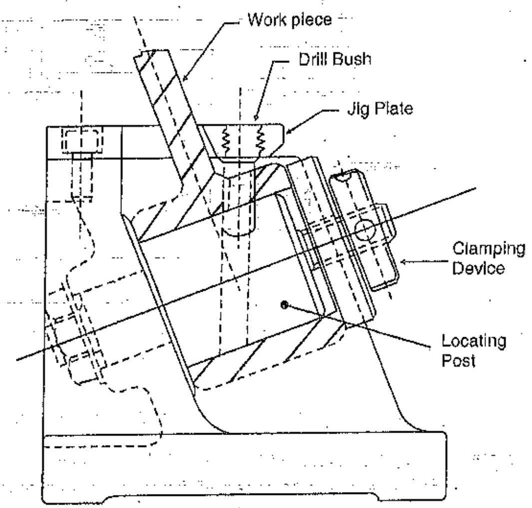


Fig. 1.50. Angular Jig

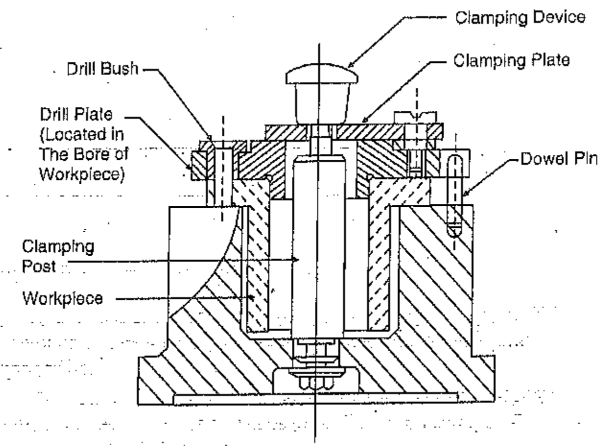
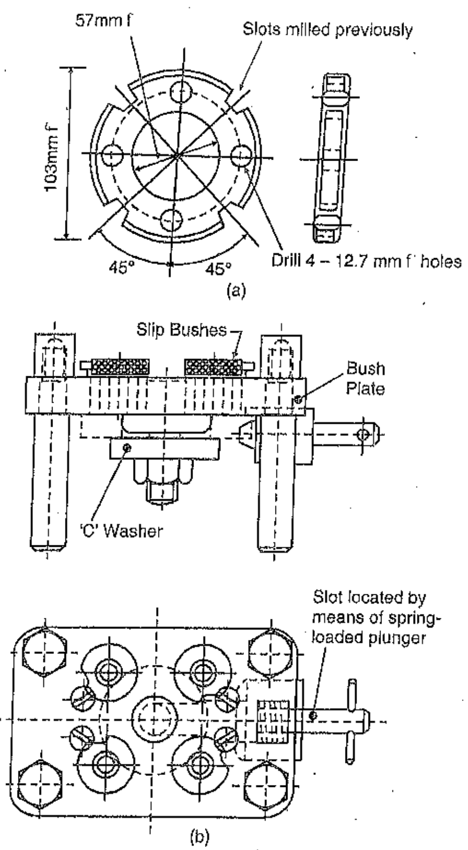


Fig. 1.51. Pot Jig

11. *Turn-over jig:* These jigs are used to drill holes in corresponding having no suitable resting surfaces. They are the modification of plate type jigs type jigs with jig feet, Fig. 1.52.



Turnover Jig For Slotted Ring
(a) Slotted ring (b) Arrangement of drilling jig

Fig. 1.52. Turn over Jig

The jig feet provide the resting surfaces to the jig during drilling. All such jigs should be "turned over" for loading and unloading the workpiece.

12. *Diameter jigs:* These types of jigs provide a simple means to drill or ream radial holes on a diameter of cylindrical or spherical jobs. The job can be very conveniently located on a V-block and clamped by a clamping plate and a clamping bolt, Fig. 1.53.

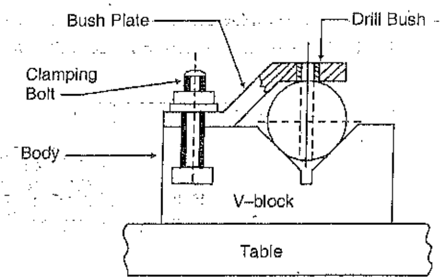


Fig. 1.53. Diameter Jigs

13. *Multi-station jig:* This type of jig has a circular indexable plate mounted on a circular base plate which is clamped to the machine table with the help of T-bolts and nuts. Such jigs are used on multi-spindle machines. Supposing the indexable circular plate has four stations. On station 1, drilling operation in being done, on station 2, reaming and on station 3, counterboring operation is being done. Station 4 is used for loading and unloading of the jobs. After every set of operations, the circular plate is indexed by 90°.

14. *Standard jigs:* There are many components that are similar in design, but different in dimensions (cylindrical pins of same diameter, but of different lengths of cylindrical pins of different diameters but of same length/different lengths, etc.) It is sometimes possible to drill several of these different components in one jig. The jig incorporates an adjustable end locator to accommodate a variety of lengths. When a jig is especially designed and fabricated for several similar parts, it is called a standard jig. The diameter jig, Fig. 1.53, provided with an adjustable end locator can be an example of such a jig.

15. *Universal jig:* This jig is first manufactured as a basic unit (just like unit heads for machine tools) to which a number of other elements and parts can be fitted to make it suitable for a specific job. Since the same basic unit can be adopted for different work-pieces and operations, simply by making additions and alterations, hence, the name "Universal Jig".

16. *Trunnion jigs:* The manual manipulation of heavy duty box type jigs is quite inconvenient and fatiguing. So, such jigs are mounted on trunnions to bring the different faces of the workpiece to the correct locations for drilling the holes.

17. *Ring jig:* Such a jig is suitable for drilling holes in round jobs, such as, flanges of pipes.

18. *Solid jig:* If holes are to be drilled in workpiece of simple shape and relatively of smaller size, then the jig body can be fabricated from a standard section of rolled steel. Such a jig can be named as a "solid Jig".

1.4.6. Jig Feet. Jig feet are needed to support the jig on the table of the machine. Jig feet should be provided opposite each working face of the jig. In normal design, the number of jig feet should be four. This is due to the reason that if any swarf or foreign matter gets under one of the jig feet, the jig with four feet will rock about this point. This will at once indicate that the jig is not resting properly on the machine table. This defect will go undetected if three feet are provided since any plane will pass through three points. So, when a jig with three feet is placed on the machine table, it will rest in almost any normal position even though swarf or some foreign matter be present under one or more of the feet. Such a jig will not be in perfect alignment with the machine spindle and the workpiece will be drilled or machined incorrectly. Hence a jig should have four feet.

Jig feet may be either cast integral with the body of the jig or fixture or built up. Cast feet are shown in Fig. 1.54. The built up feet (Fig. 1.55) can be installed into the jig body in many

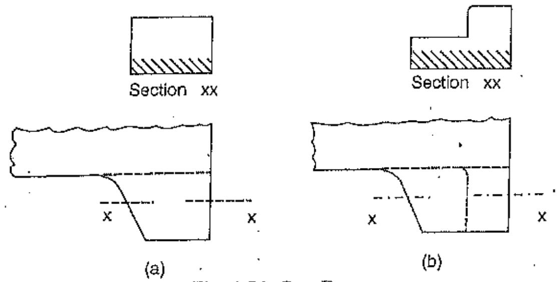


Fig. 1.54. Cast Feet.

ways. In Fig. 1.55 (*a*) the foot is press fitted into the jig body. A drift hole should be provided for forcing the foot out. In Fig. 1.55 (*b*) the foot is screwed into the jig body. Figs 1.55 (*c*) and 1.55 (*d*) show turnover feet. The short foot is used to support the jig when it is turned over.

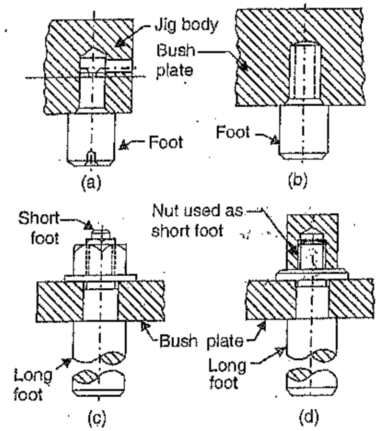


Fig. 1.55. Built up Feet.

In both cases, jig feet are held by a shoulder on the jig feet against the jig body by a nut. The supporting faces of the steel feet should be hardened and finished ground.

A drilling jig for machining a hole in a casting is shown in Fig. 1.56.

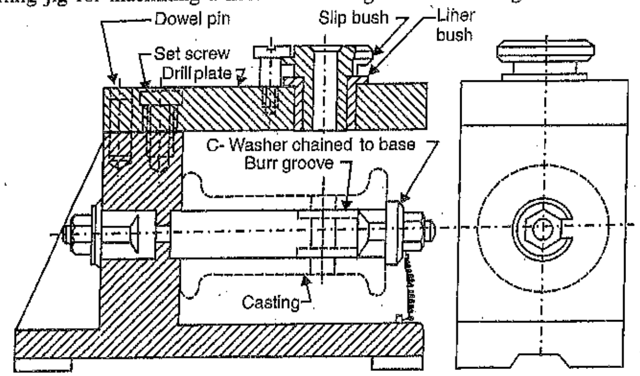


Fig. 1.56. A Drilling Jig.

1.5. MILLING FIXTURES

A milling fixture is a work holding device which is firmly clamped to the table of the milling machine. It holds the workpiece in correct position as the table movement carries it past the cutter or cutters. A milling fixture is usually a good solid casting, since cast iron has the important characteristic of absorbing and damping out the vibrations resulting from the cutting action of the milling cutters. The essential features of a milling fixture are : A heavy base, location and clamping elements and setting blocks.

(*i*) *Base* : A heavy base is the most important element of a milling fixture. It is a plate with a flat and smooth underface. The complete fixture is built up from this plate. Keys are provided on the underface of the plate which are used for easy and accurate aligning of the fixture on the milling machine table by inserting them into one of the *T* slots in the table. These keys are usually set in keyways on the undersurface of the plate and are held in place by a socket head cap screw for each key Fig. (1.57). The fixture is fastened to the machine table with the help of two *T* bolts engaging in the *T*-slots of the table and protruding through the hold down slots or ears in the mill fixture base.

(*ii*) *Location and clamping elements.* The same design principles of location and clamping apply for milling fixtures as have been discussed earlier.

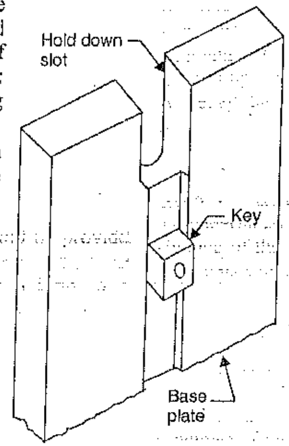


Fig. 1.57. Milling Fixture Base.

(*iii*) *Setting blocks.* After the fixture has been securely clamped to the machine table, the workpiece which is correctly located in the fixture, has to be set in correct relationship to the cutters. This is achieved by the use of a setting block and feeler gauges. The setting block is fixed to the fixture. Feeler gauges are placed between the cutter and reference planes on the setting block so that correct depth of cut and correct lateral setting is obtained (Fig. 1.58). The block is made of steel, hardened and with the reference planes (feeler surfaces) ground. Sometimes, if convenient, the feeler surfaces may be machined in a part of the fixture. These should be clearly marked so that the operator is certain as to which are the feeler surfaces. In its correct setting, the cutter should clear the feeler surfaces by at least 0.08 cm to avoid any damage to the block when the machine table is moved back to unload the fixture. The thickness of the feeler gauge to be used should be stamped on the fixture base near the setting block. It is common to keep a gap of 0.4 to 0.5 mm (0.8 mm noted above) between the cutter and the setting planes of the setting block. This gap facilitates the cutter setting with the help of feeler gauges, Fig 1.58.

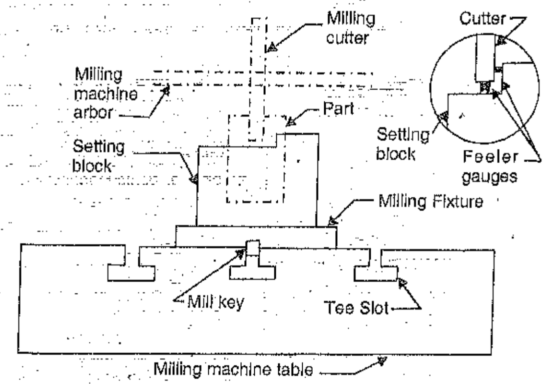


Fig. 1.58. Setting Block.

1.5.1. Milling Machine Vice. For many jobs, the standard machine vice can be used to clamp the workpiece under cutter, Fig. 1.59. The movable jaw can be operated either by a cam or a screw. To suit the contours of the

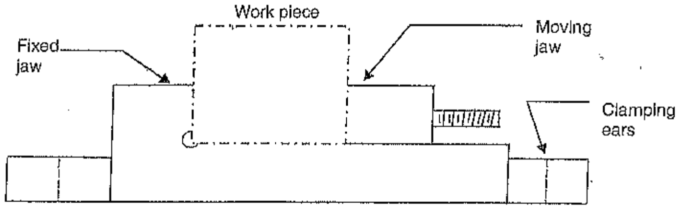


Fig. 1.59. Milling Vise.

various jobs, special jaws may be attached to the plain jaws of the standard vice, and vees may also be fitted instead of the standard plain jaws, Fig. 1.60.

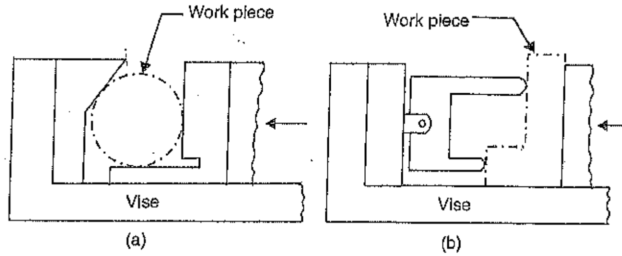


Fig. 1.60. Special Jaws for the Vice.

For short runs, the special jaws may be made of aluminium alloy or soft mild steel. For long runs, the wearing surfaces of the special jaws should be hardened and should be of renewable type. Cast iron jaws are also frequently used. Wherever possible and convenient, setting blocks should also be provided.

1.5.2. Some Design Principles for Milling Fixtures. 1. Pressure of cut should always be against the solid part of the fixture, Fig. 1.61.

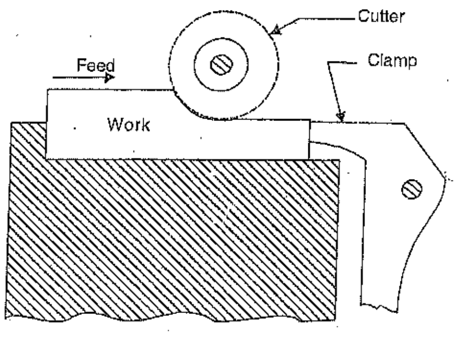


Fig. 1.61.

2. Clamps should always operate from the front of the fixture, Fig. 1.62.
3. The workpiece should be supported as near the tool thrust as possible, Fig. 1.63.

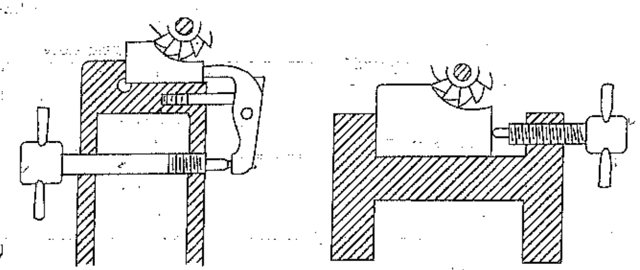


Fig. 1.62

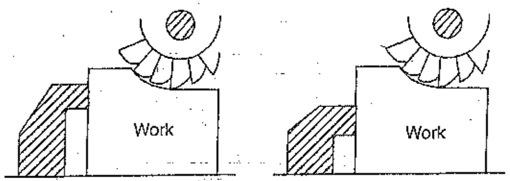


Fig. 1.63

1.5.3. Classification of Milling Fixtures. There is no standard method of classification of milling fixtures. This is because, each fixture is different and must be designed to meet certain requirements. However, these may be classified in the manner discussed below:

1. **According to the type of the milling operation performed on the work.** According to this criteria, the milling fixtures can be : Face milling fixtures, slab milling fixtures, slotting fixtures, straddle milling fixtures, gang milling fixtures, string or line milling fixtures, profile milling fixtures and so on.

2. **According to the way the workpiece is clamped.** Under this category, we have : Hand clamping fixture, power clamping fixtures, toggle fixtures or automatic fixtures.

3. **According to the way the workpiece is located.** Such as centre fixture, V-block fixtures and pin or stud fixtures.

4. **According to the method of presenting the workpeice to the cutter.** Here we have:

(*a*) *Cradle Fixtures.* Workpiece is rocked or rotated within a given angle during milling.

(*b*) *Rotary Fixtures.* Workpiece is rotated under the cutter.

(*c*) *Drum Fixtures.* Workpiece is mounted on the periphery of a rotating drum.

(*d*) *Indexing Fixtures.* Workpiece is indexed into the next position during the machining cycle of the mill.

(*e*) *Rise and Fall Fixtures.* These fixtures allow raising and lowering of the workpiece in conjunction with the mill feed.

5. **Universal fixture.** These fixtures are designed to hold a family of parts similar in shape but different in size.

6. **Progressive fixtures.** Here the workpiece can be located in different positions and progressively moved from one fixture station to the next until all operations are complete.

7. **Temporary fixtures.** These fixtures are built of standard clamps, T-bolts, heel blocks, jacks and stops and are used in tool room operation and small lot production.

8. **Fixtures for higher rate of production.** Here, the fixtures are frequently used in pairs. Two identical fixtures mounted on opposite ends of a milling machine table, permit the machinist to load one while the other is in operation.

9. **Fixtures for high production.** Rotary table attachment is mounted on the table and connected to the feed mechanism of the machine in such a way that it will rotate slowly. Several fixtures are mounted on the rotary table, Fig. 1.64, and a part is clamped in each. As a part is finished, the operator merely removes it and inserts an unfinished one. With this set-up, no cutting time is lost during loading and unloading, because, the rotary table turns continuously. The method is widely used for face milling, but can be used for slotting also.

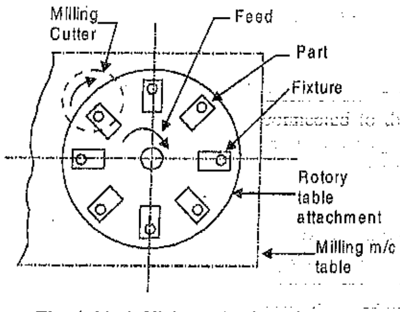


Fig. 1.64. A High Production Fixture.

10. **Special vise jaws.** See Fig. 1.60.

1.6. LATHE FIXTURES (TURNING FIXTURES)

The standard workholding devices or fixtures for a lathe are: three and four jaw chucks, collets, face plate, mandrels and milling vice. Three jaw chucks are used to hold round or hexagonal bar stock or other symmetrical work. Four jaw chucks are used for rough castings and square or octagonal work. When irregular shaped castings and forgings are to be machined, special chuck jaws may be used to suit the component. Collets are used primarily for bar stock. Some types of jobs can be mounted and supported on a face plate. Standard equipment such as angle brackets and flat plates may be screwed or bolted to the face plate and the work clamped to these. For example, jobs which are hard to hold such as pipe elbow may be bolted to an angle plate which in turn is bolted to the face plate. Hollow jobs such as gear blanks and pulleys can be supported on a mandrel between centres and rotated by a dog. A mandrel is a hardened bar held between centres. It has flats on each end for clamping a lathe dog. The mandrel is tapered about 0.5 mm per metre so that the job can be forced on it with a press fit and then unlocated after machining. A milling vice can be mounted on the lathe bed in place of the compound rest, to hold the work and the cutter is inserted in the lathe spindle.

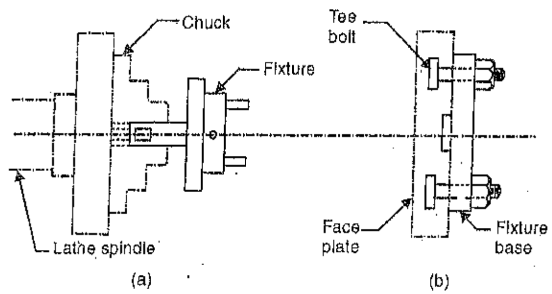


Fig. 1.65. Fixing of a Lathe Fixture.

If the job can be held easily and quickly in the above mentioned standard devices, then there is no need for a special workholding device. However, many jobs, particularly castings and forgings, because of their shape, cannot be conveniently held by any of the standard devices. It then becomes necessary to build a special work holding device for the job. Such a device is called a lathe fixture. A lathe fixture consists of a base, location and clamping devices as usual and an arrangement for locating and fastening the fixture securely and accurately in the lathe. A lathe fixture can be fixed to the lathe either by holding in the chuck jaws or fixing to a face plate, Fig. 1.65. The following design points should be considered for a lathe fixture:

1. To avoid vibrations while revolving, the fixtures should be accurately balanced.

2. There should be no projections of the fixture which may cause injury to the operator.

3. The fixture should be rigid and overhang should be kept minimum possible so that there is no bending action.

4. Clamps used to fix the fixture to the lathe should be designed properly so that they don't get loosened by centrifugal force.

5. The fixture should be as light weight as possible since it is rotating.

6. The fixture must be small enough so that it can be mounted and revolved without hitting the bed of the lathe for which it is designed. The designer should check the "swing" of the lathe.

7. The part must be mounted and held in such a way that there will be no obstructions in the way of the cutting tool or tools.

1.7. GRINDING FIXTURES

The workholding devices for grinding operations will depend upon the type of grinding operation and the machine used. We shall take each case one by one, as below :

1. **Fixtures for external grinding.** A mandrel is the most common fixture used for grinding external surface of a workpiece. As written earlier, a mandrel is hardened and is held between centres of a machine. The mandrel is used for internal chucking of round workpiece with bores. The workpiece is located and held on the mandrel with the help of the bore so that the external surface may be machined truly concentric to the bore. The various types of mandrel are:

(*i*) *Taper mandrel.* In this type of mandrel, the outer chucking surface is given a slender taper of about 0.5 mm per metre, Fig. 1.66. The diameter of the bore of the workpiece must be smaller than the largest diameter of the mandrel. The workpiece is forced endwise on to the mandrel. The gripping force between the workpiece and the mandrel is created due to the taper on the mandrel.

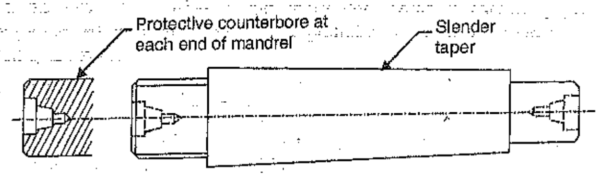


Fig. 1.66. Taper Mandrel.

(*ii*) *Straight mandrel.* It differs from the taper mandrel in that it has straight or untapered chucking surface. To press fit the workpiece on to the mandrel, the outer diameter of the mandrel is made bigger than the bore of the workpiece. The difference known as interference depends upon the wall thickness, diameter and the material of the workpiece. It should not be able to cause plastic deformation of the mandrel and/or of the workpiece.

(*iii*) *Combined taper and straight mandrel.* In this type of mandrel, a portion of the outer diameter of the mandrel is straight and the rest is tapered. The straight portion helps to prealign the bore of the workpiece and the tapered portion provides the driving area on to which the workpiece is press fitted.

2. **Fixtures for internal grinding.** For grinding internal surfaces of simple circular workpieces, the chuck may be used as a standard workholding device. If required, special jaws can be provided for the chuck. However, for many components, special fixtures may have to be made which are designed on the same lines, as the lathe fixtures.

3. **Fixtures for surface grinding.** The workpiece can be held for machining on a surface grinder in the following ways:

(*i*) It may be clamped directly to the machine table or to an angle plate and so on.

(*ii*) It may be held in a vice.

(*iii*) The workpiece may be held by means of a magnetic chuck or a vacuum chuck. Here, the workpiece is held without any mechanical clamping.

(*iv*) The workpiece may be held in a special fixture.

(*a*) **Magnetic chuck.** A magnetic chuck holds a workpiece by the magnetic force which may be generated either by permanent magnets or by electromagnets powered by direct current. The gripping power of the chuck depends on the strength of the magnets and amount of magnetic flux which passes through the workpiece. The magnetic chuck is clamped to the machine table. Magnetic chucks are available in various shapes such as rectangular, circular and also as a *V*-block. A magnetic chuck has the advantage of being fast acting and causing minimum distortion of the workpiece. However, it can be used for holding only ferrous workpieces. Also, some residual magnetism is imparted to the workpiece which must be removed by demagnetising it.

(*b*) **Vacuum chucking.** This is a very convenient method of holding workpieces of non-magnetic materials. The principle of vacuum chucking is explained in Fig. 1.67. The workpiece is placed on the '*O*' ring seal inserted in the workholder. This forms a close chamber between the locating surface of the workpiece and the mating surface of the workholder. At start, the air pressure on the inside and outside of the chamber will be equal. Then a vacuum is created in the chamber. The outside pressure then holds the workpiece against the locating surface of the workholder.

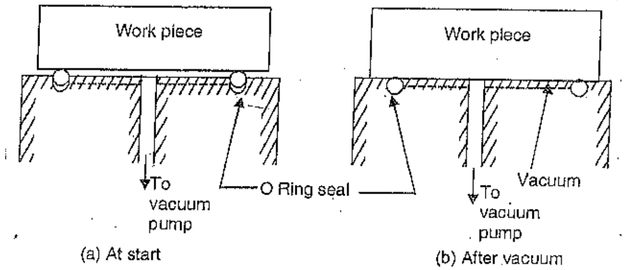


Fig. 1.67. Vacuum Chucking.

1.8. BROACHING FIXTURES

Broaching fixtures are must for workholding because of the high forces involved and because of the manner in which the operation is performed. Broaching fixtures are required to perform one or more of the following functions:

1. Hold the job rigidly.
2. Locate the job in correct position relative to the tool of the machine table.
3. Guide the broaching tool in relation to the job.

4. Move the job into and out of the cutting position.
5. Index the job between the cuts.

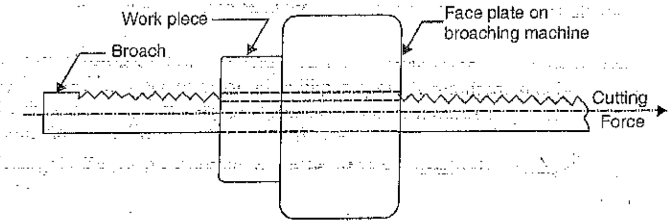


Fig. 1.68. Broaching Fixture.

Fixtures are needed for internal broaching and external broaching. The fixtures used for internal broaching are the simplest and for many operations may consist of a face plate or support plate on the broaching machine (Fig. 1.68). The fixtures for external broaching are made quite rigid so that the workpiece does not move during the broaching action.

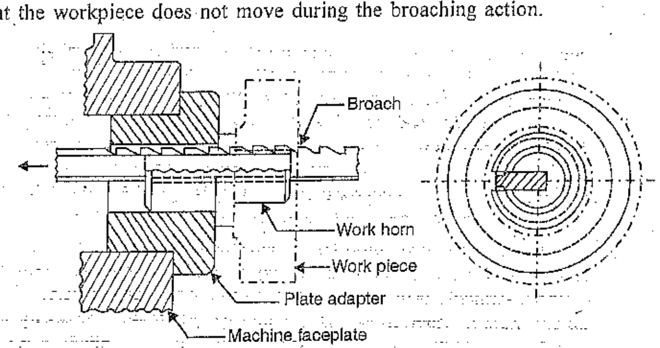


Fig. 1.69. A Simple Keyway Broaching Fixture.

A simple broaching fixture used to machine a keyway is shown in Fig. 1.69. The fixture is simply a plug or horn fixture which establishes the correct position of the workpiece in relation to the broach and the machine faceplate. The broaching force will hold the work in position of the fixture. Special work horn may be designed to broach keyways at an angle, Fig. 1.70.

In the case of external broaching, the broaching force will not hold the work in position on the fixture, but will push it away, Fig. 1.71. Therefore, surface broaching fixtures are more elaborate and require proper clamping arrangements. External broaching will usually require a special fixture for each job.

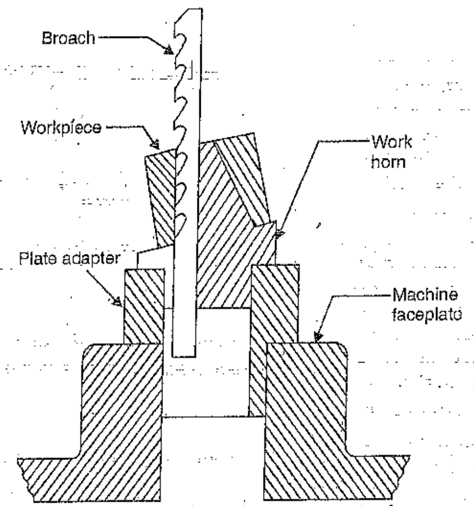


Fig. 1.70. Broaching Keyway at an Angle.

1.9. ASSEMBLY FIXTURES

Assembly fixtures are used to hold the various components in their correct position while they are assembled. This is particularly the case when the various parts are to

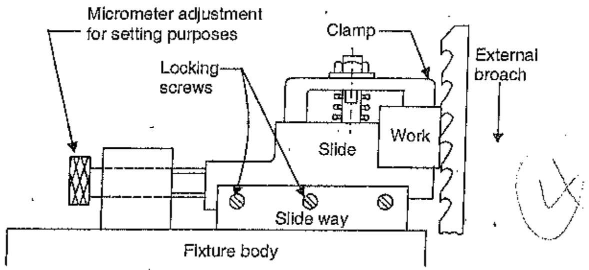


Fig. 1.71. External Broaching Fixture.

be put together for welding. They are indispensable in the component and final assembly of units. They are simple and effective means used to mechanize the manual assembly operations. Their use assures higher productivity and convenient performance of the assembling operations, securing a quick and accurate fixturing of the mating parts during assembly. According to the degree of specialization, they are classed into : Universal, and Special-purpose fixtures.

The 'universal fixtures' are used in single-piece and small-lot production. These include vices of all types, match plates, pedestal bars, V-blocks, angles, crampling frames, lifting jacks, and various other ancillary hold devices (rests, or stands, wedges, screw clamps, etc.). Match plates and bars serve as basements in bench assembly to mount, align and clamp machine assemblies and components and are usually made of cast iron. Angles and V-blocks are used to locate and clamp the components and positioning pieces. Usually they have machined locating surfaces and longitudinal through holes to receive fastening bolts. Clamping frames are used for a temporary holding of parts and units to perform certain auxiliary operations like straightening, pressing-in, pressing-out, etc. Jacks are used as adjustable supports in aligning large-size and heavy parts and assemblies before they are put together.

The 'special-purpose fixtures' are used in quantity production for performing special assembly operations. They can be ; Static (work-benches and assembly stands) and mobile which are fixed to plates of chain conveyors, turntables, cars, dollies, etc.

The static fixtures are set up-clamping or fit-up fixtures used to fix in position and clamp the adjuster or positioning pieces and companion parts of the product assembly. Fixtures of this type give the adjuster the necessary degree of stability during the assembly operation, raise productivity because the operator has his both hands free to do the assembling.

For convenience and higher productivity, fixtures are often made turnable.

The workholders intended for fixturing base members and component parts of assemblies can be : Unitary and Multiple. Unitary fixtures are used to clamp one product assembly. The application of multiple fixtures raises the productivity of assembling due to the reduction of the auxiliary time for workloading. Assembling in a multiple fixture operates on the principle of consecutive or parallel concentration of manufacturing steps. An example of the later can be the simultaneous tightening of threaded joints in all parts, located in the fixture, with the help of a multiple-spindle nut-driver. Multiple fixture must assure a uniform and quick grasping of all the parts of assembly, for example, with an air operated clamp. Such fixtures can be static and mobile. The static fixtures are mounted in workbenches or assembly stands and the mobile, in cars or plates of conveyors.

Another type of special-purpose assembly fixtures includes work holders designed specifically for quick and accurate fixturing of mating parts and components of a product assembly. The assembly fixtures of this type free the operator from the need to check and align the relative position of the mating parts because accurate positioning is achieved automatically

by bringing these parts into positive contact with the fixture supporting and guiding elements. Such fixtures are frequently used in welding, brazing and soldering, riverting glueing, expanding, fitting parts with interference into threaded and other assembly joints. These fixtures assure a considerable rise in productivity and are indispensable in automatic assembly. The special-purpose assembly fixture consists of a body (base member) on which mounted are the locating elements and clamping devices. The fixture need only be of light construction with adequate rigidity to ensure relative positional relationships of the various components. They may be built-up from light castings, steel sections or completely from steel.

Some simple types of welding fixtures are shown in Fig. 1.72. The parts to be welded are placed in the fixture in their correct relationship and the whole arrangement tightened. The parts are then welded whilst in place. The locating and clamping principles described earlier, also apply to welding fixtures. However, a few design principles applicable only to welding fixtures are given below :

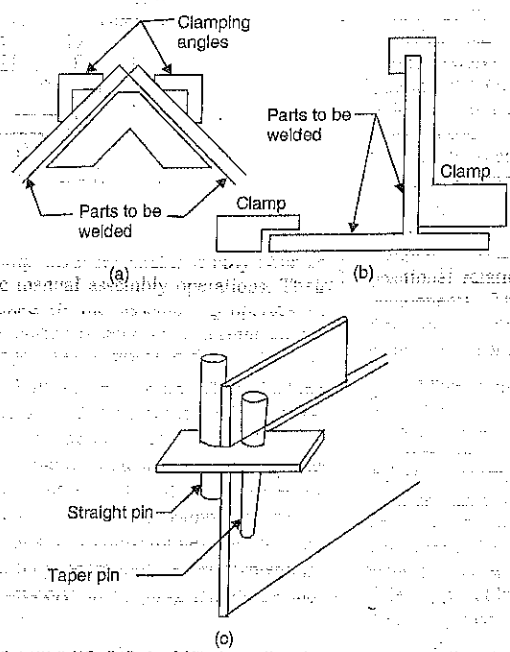


Fig. 1.72. Simple Welding Fixtures.

1. Due to intense heat produced during welding process, expansion will take place. Due to this, the welded assembly may get locked in the fixture, making its removal very difficult. To avoid this, the locators should be designed accordingly.

2. Clamping pressure should be light and clamps should be arranged in such a way that the workpiece does not get distorted.

3. Clamps should be kept clear of welding zone or be shielded.

4. Location and clamping should not make the welding zone inaccessible.

5. The welding fixture should be rigid and stable.

1.10. INSPECTION FIXTURES

An inspection (qualifying, gauging) operation is any examination of a workpiece that determines whether or not it meets the standards of quality. Inspection fixtures are used to check the quality of the workpieces, parts and components of machines. Dimension inspection or guaging fixtures raise the efficiency of the work of human inspectors, improve their working conditions, quality of workpieces, parts and components of machines. Fixtures for checking parts are usually employed at intermediate stages of machining (step-by-step inspection) and at the final stage of machining (acceptance inspection) to verify the accuracy of dimensions, relative position of surfaces and adequacy of surface geometry.

High precision of modern machines is conducive to the use of inspection fixtures of high sensitivity meters and gauges. Small and medium-size parts are checked in stationary fixtures and large-size parts and products, in portable fixtures. Gauging fixtures can be single-and multi-limit control devices. The last are used to measure several parameters in one set up.

Inspection fixtures are classed into in process and off-line gauging attachments. The latter are used for post-operation inspection. The in-process inspection attachments are mounted on machine tools to control the dimensions of parts during machining by signalling the machine control devices or the operator to stop machining or change the cutting conditions as the controllable variable goes off limit. In that way, independent inspection fixtures become an integral part of the automatic control system. This permits cutting down the cost of production by eliminating rejects and separate quality control operations.

Dimension inspection fixtures are designed to secure the pre-set accuracy and efficiency of the quality control operations and must be convenient in use, simple in construction, cheap and reliable in service. These fixtures need not be designed to withstand forces, such as shock and vibration, associated with machining or with some other fabricating and assembly processes. They are not required to resist temperatures present in welding, brazing etc. Clamping forces in an inspection fixture are generally too small to affect its design, but they should not distort the workpiece.

Elements of Inspection Fixture. The inspection or gauging fixture consists of locating, clamping, gauging and auxiliary elements mounted in the body of the fixture. The locating elements (supports) are used to locate the part or workpiece from its reference for gauging. The clamping devices prevent displacement of part (assembly) set in it for checking relative to the gauging device and assure positive contact between the part's locations and fixture's supports. The performance of the clamping devices in inspection fixtures differs greatly from that of clamping devices in machining attachments. To prevent deformation of the inspected product, the former should develop small clamping forces that remain stable with time. Clamping devices are not needed if the part is quite stable in the fixture's supports and the gauging forces do not disturb its stability in the fixture. To enhance the efficiency of inspection, it is advisable that the clamping device be quick-grasp and convenient in use.

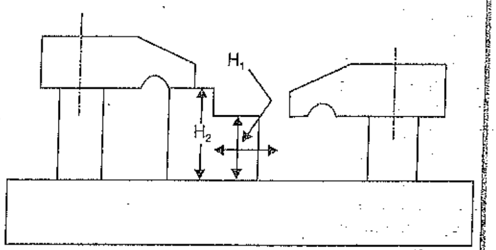


Fig. 1.73. An Inspection Fixture with Rigidly Fixed Limit Elements.

The clamping devices in inspection fixtures are usually hand-operated (lever and spring-actuated, screw and eccentric-type clamps) and power devices, such as pneumatic clamps whose power is also used to actuate the fixture's ancillary mechanisms (lifters, rotators, pushers).

Application of multiple gauging fixtures enhances the quality of control, frees human labour and reduces the cost of instrumentation. In order to make multiple gauging applicable, it is necessary that the controllable dimensions be measured against a common reference location.

The measuring instruments in inspection fixtures can be control-limit (non dial) and reading (dial) gauges. The devices which employ the principle of normal gauges constitute a special group of size-control instruments.

The simplest fixture arrangement is one which has rigidly fixed limit elements to check the height of the shoulders (dimensions H_1 and H_2) of a stepped part being moved along the baseplate of the fixture by hand, Fig. 1.73, the lower surface being the reference surface. The limit elements can also be extensible.

1.11. BORING FIXTURES

In boring operation, an already existing hole in the workpiece is enlarged. The boring tool is usually a single point tool and the size of the tool will depend upon the adjustment of the tool within the boring bar. The boring operation may be done by either of the two arrangements–boring bar is stationary and the workpiece moves into the bar or the workpiece is stationary and the boring bar moves into the workpiece. Boring fixtures can be divided into two general classes: one, where the fixture guides the boring bar as in drill jigs and it is more appropriately called a Boring jig, and the other in which fixture holds the workpiece in proper relation to the boring bar, as in milling fixture. The various machine tools employed for boring operation can be : lathes, drill presses, milling machines, jig boring machines and vertical or horizontal boring mills. According to its length, a boring bar may be classified as : a stub bar, single piloted bar and double piloted bar. A stub boring bar is short in length and is not supported along its length. It is used for boring short holes and also blinds holes. A single piloted bar is longer than a stub bar and is supported at its leading end so that it does not spring under cutter thrust, Fig. 1.74. A double piloted bar is supported both at leading and trailing ends, Fig. 1.75. Sometimes, slip bushings fitted into the finished holes in the workpiece may act as support and guide for the boring bar, Fig. 1.76. In order to eliminate any trouble from misalignment of the jig and the spindle of the boring machine, the boring bar is connected to the spindle by a short intermediate shaft fitted with a universal joint at each end. The general principles of jig and fixture design are also applicable for boring jigs and boring fixtures.

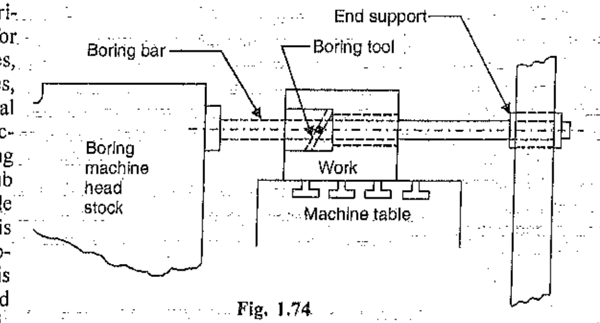


Fig. 1.74

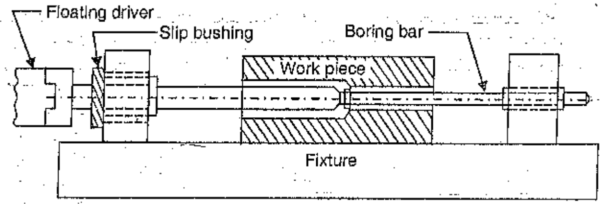


Fig. 1.75

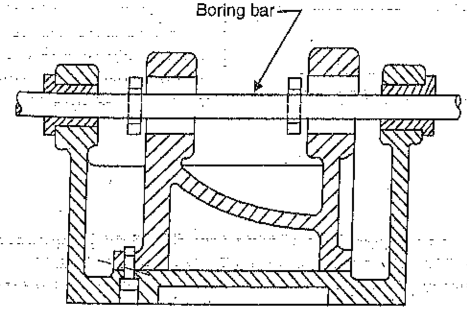


Fig. 1.76

1.12. PLANING AND SHAPING FIXTURES

We know that Planers and Shapers are not very efficient machine tools. Due to this, they are usually not recommended for mass production. However, jigs and fixtures are normally used on machine tools engaged in mass production. As such fixtures do not find much utility on Planers and Shapers. Most of the time, the general purpose work-holding devices are used on these machine tools; such as, T-bolts and clamps, stop pins and Toe dogs and Vises (See the book by the author: A T.B.O. Production Technology). However if a job can not be held in one of these devices, then a fixture will have to be designed and fabricated.

1.13. INDEXING JIGS AND FIXTURES

Indexing jigs and fixtures are used when holes or slots are to be machined to some specific relationship, in a workpiece. These are of particular interest for drilling, milling, and grinding operations. There are two ways in which a workpiece may be moved under the cutting tool for machining each hole or slot or for other operation:

(*i*) By rotating the workpiece.

(*ii*) By a sliding movement of the workpiece.

The first method is employed for drilling, milling and surface grinding operations, whereas the second method finds application on production lathes and on external grinding machines.

1.13.1. Indexing devices. Many indexing jigs and fixtures employ a simple indexing plate for their operation (Fig. 1.77). Suppose six holes are to be drilled in a flange. The flange can be mounted on an index plate which has six equispaced slots. The workpiece is revolved under the drill and each hole is drilled in turn. For this an index plunger is used which fits by turn into each slot in the index plate. To index the workpiece, the plunger is pulled out of the slot. The index plate and thereby the workpiece is rotated until the next slot comes in line with the index plunger into which it gets pushed due to spring action. Some other indexing devices involving rotation of the workpiece are shown in Fig. 1.78. In Fig. 1.78 (*a*), a spring loaded steel ball is employed. For indexing, the rotating member is revolved by hand. Due to the turning effect, the ball is pushed out of the groove in the rotating member, which becomes free to revolve. It is rotated until the next groove in the rotating member is encountered by the ball into which it gets pushed due to spring action. This device is not very accurate. In Fig. 1.78 (*b*), a steel peg is used which is retained in the fixed member by means of a key and keyway. For indexing, the peg is pulled out clear of the bush in the rotating member. The rotating member is then revolved until the next bush is encountered by the peg, into which it passes.

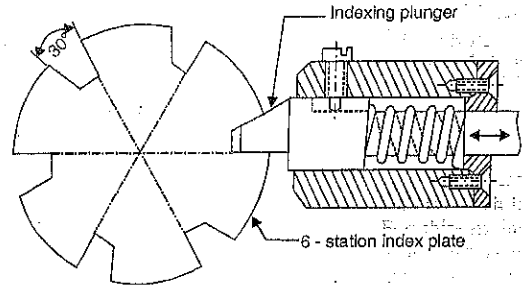


Fig. 1.77 Indexing Plate

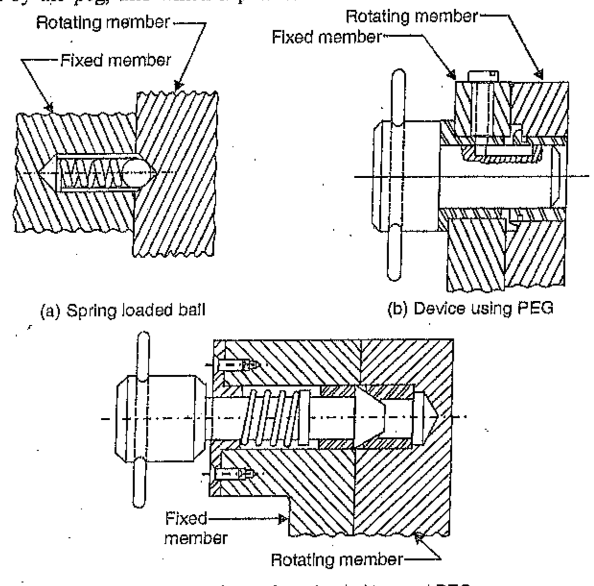


Fig. 1.78. Indexing Devices.

The same principle is employed in the device shown in Fig. 1.78 (*c*). Here the peg is spring loaded. The arrangement using a sliding member is shown in Fig. 1.79. The sliding member is provided with slot at suitable spacings and the fixed member carries a spring loaded lever, which fits the slots. For indexing, it is lifted off a slot and the sliding member is moved until the next slot comes under the lever.

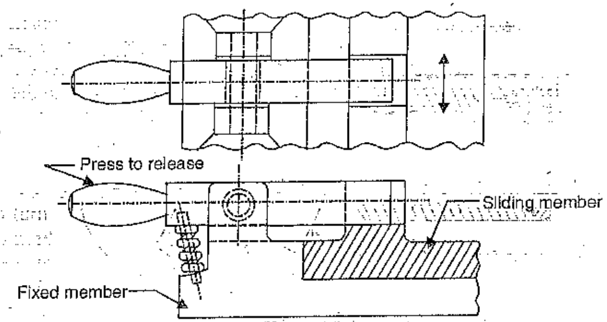


Fig. 1.79. Sliding Indexing Device.

1.14. AUTOMATED JIGS AND FIXTURES

In machining and assembling operations, the auxiliary time forms quite a large percentage of the standard time per piece. The auxiliary operations have to be done hundreds of times per day, and if done manually may cause considerable fatigue to the operator, thereby reducing his efficiency. Also, the time spent in this activity can seriously affect the production. These problems can be overcome by making the performance of jigs/fixtures automatic provided it is economically justified by the rate of production. This will assure higher efficiency and liberate the human labour considerably. Automation can be partial or full.

In 'partial automation', we automate one or several steps or passes : loading and unloading of jobs by various means : clamping and unclamping of workpieces; removal and push-out of workpieces from the workzone; rotation, indexing and locking of the rotating parts of multiple and mulit-place fixtures; in process inspection of workpieces. In 'fully automated jigs/fixtures' and machining cycles, the function of the operator is limited only to the loading of workpieces in hoppers (feeders) and control over the performance of jigs/fixtures and the machine tool.

Automated jigs/fixtures should exclude the possibility for incorrect location of workpieces. For this, they employ interlocking and safeguarding devices and clearance limits (in transfer lines) to the effect that the machine tool stops machining with the work in incorrect position (or absent) in the fixture.

The driving mechanisms in jigs/fixtures can be : (1) Mechanical (2) Pneumatic or Air operated (3) Hydraulic (4) Pneumohydraulic (5) Electrical, or a combination of these.

Air operated drives are more widely used due to the ready availability of the compressed air in most of the production shops. Their advantages are : relative simplicity and low cost, quick action and reliability in service, absence of return pipeline, insensibility to the changes of the ambient. The used air can be used again for swarf disposal and to transfer further small-sized machine parts or product assemblies. However, these drives are impracticable where large forces are required for clamping, because with pressures between 40 and 60 MPa, they have large overall dimensions and small efficiency, produce much noise by discharging the used air, and fail to assure smooth running of the fixture's working parts. Pneumatic drives are essentially air cylinders (linear piston actuators).

Hydraulic drives are used particularly with relatively large components and on hydraulic machine tools. They have the following advantages :

1. They are relatively small and compact due to high working pressures involved (400 to 600 MPa and higher).

2. As the working fluid is oil, there is no need of external lubrication. Also, there is no trouble of water condensation which may be there in a pneumatic system.

3. Since the working fluid is almost incompressible, positive action of the operating parts of the components is ensured.

4. They are noiseless, smooth, have higher efficiency (upto 80-90%) and small delay time (0.01-0.02s).

Their disadvantages include the presence of individual or general-purpose tank of sufficient capacity, a pump, an oil drain line, sensitivity to the variation of oil viscosity at heating, and a high cost. Hydraulic drives are essentially linear piston actuators (hydraulic cylinders) operated by a separate pump. The air-operated and hydraulic drives are capable of withstanding overloads and are easy to automate.

Pneumohydraulic drives have small-sized actuators, normally air-operated. They ensure quick motions of idle and auxiliary steps and passes, automatic switching over to the manufacturing steps with a required decleration of the speed of the machine working part.

Compared to hydraulic drives, the electric drives have a still higher speed of response, low power consumption, larger overall dimensions and mass, higher sensitivity to overload and overheating, and lower reliability in service. However, they have higher efficiency as compared to pneumatic and hydraulic drives.

Drives are controlled by cams and stops, master controllers, servo-valves and limit switches that operate in response to the action by the machine moving parts (tables in millers and spindles in boring and drilling machine tools).

1.14.1. Automatic Clamping Devices. The following advantages accrue from the use of automatic clamping devices :

1. Clamping is quick, thereby, considerably reducing the chucking time.

2. The clamping force is uniform and is independent of any personal variation caused by the operator.

3. More clamping force is obtainable than in manual clamping.

4. Two or more clamps can be operated simultaneously from a single control print.

5. As the working pressures are high, the automatic clamping devices (pneumatic and hydraulic) are small and compact.

In the simplest case, automatic clamping is achieved by applying clamps actuated by the feed mechanisms or cutting forces of the machine tool itself. Feed-actuated clamping devices are used largely in drilling machines. In a multi-spindle drilling machine, the jig plate is located in between the spindle head and the table and is supported on the pillars on which the spindle head is supported. The jig plate is spring loaded. As the spindle head is lowered, the jig plate moves towards the table. Subsequent lowering of the spindle head compresses the springs. This causes the clamping force to grow steadily to reach its maximum by the end of drilling. The drawback is the additional loading of the machine feed mechanism.

Electromagnetic, magnetic and vacuum clamping devices have already been discussed under 'Grinding Fixtures'.

Motor driven clamping devices, usually find applications in machine tools of the turret-lathe group, in transfer machines and transfer lines. They have screw clamps driven by a wrench which is a withdrawable device (manually or automatically) with a torque limiting sprial jaw clutch. Torque from the motor shaft is transferred through a reducer and a clutch to a screw which drives a nut linked to the rod of the clamping device. Upon reaching the desired clamping force, the clutch disengages. The torque is controlled by adjusting the strength of the torque-control spring. Unclamping is affected by reversing the motor, Fig. 1.80.

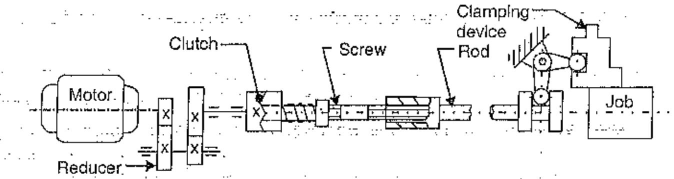


Fig. 1.80. Motor-driven Clamping Device

For pneumatically controlled clamping devices, compressed air is needed. For this, the installation includes a compressor, air receiver and a pipe line around the shop. Coupling points at suitable intervals are provided where a pipe may be plugged in. The working pressure is usually about 0.3 to 0.6 MPa. The operating equipment consists of a piston and cylinder arrangement with the piston rod connected to the levers or to some other mechanisms which comprise the method of clamping. To clamp and unclamp a workpiece, the piston must operate in both directions. For this, air inlets are provided at each end of the cylinder and a valve is provided to regulate the supply of air. The various arrangements of air operated clamping devices are shown in Fig. 1.81. The arrangements [Fig. 1.81 (*a*), (*b*)] are suitable where slight variations may be encountered since the toggel and cam clamp will remain securely locked under pressure.

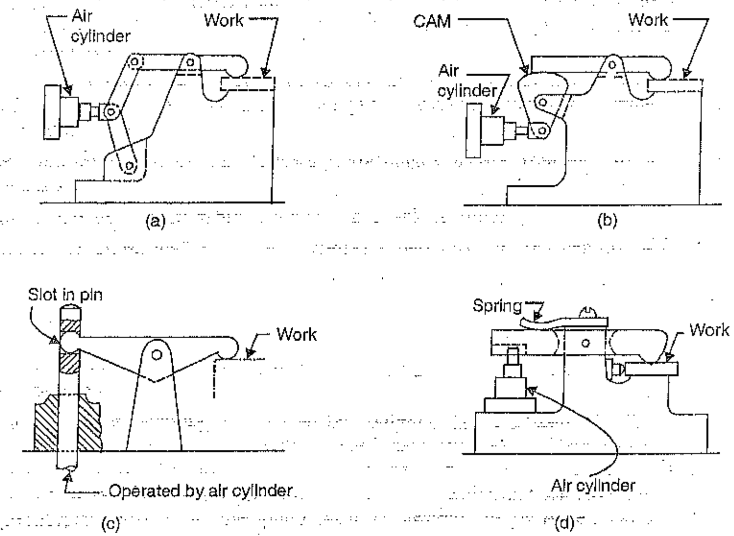


Fig. 1.81. Air Operated Clamping Devices.

Hydraulically operated clamps operate similarly to air operated clamps.

1.14.2. Rack and Pinion operated Clamping Device. Rack and pinion clamps, Fig. 1.82, consists of a rack, pinion mounted on a shaft and a handle. By turning the handle counterclockwise, the rack gets lowered resulting in the clamping of the job directly or through an intermediate piece (e.g., a plate). The clamping force depends on force F applied to the handle. To retain the developed clamping force after removing the applied force, the clamp has a lock to prevent the reversal of the rack pinion under the action of elastic forces due to arise in the links of the clamping system. For unloading the job at the end of the machining operation, the handle is turned clock-wise.

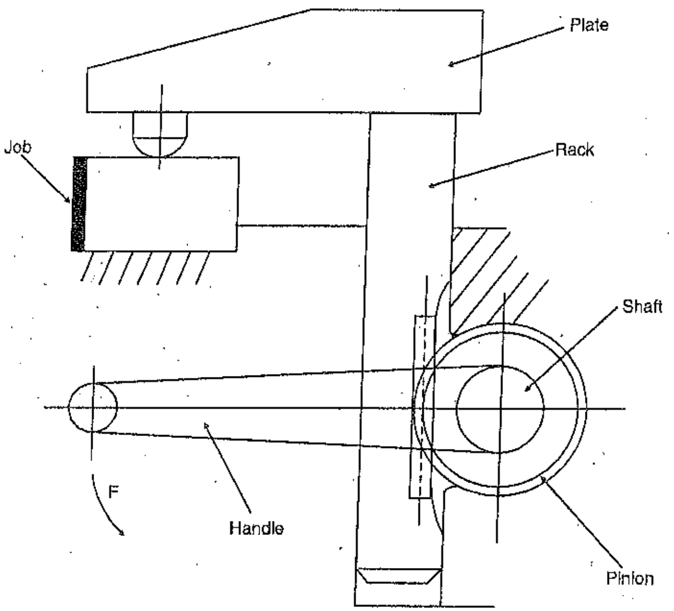


Fig. 1.82. Rack and Pinion Clamp.

Such a clamping device is extensively used on universal jigs. The rack and pinion clamping device is not self-locking (a pressure on the rack will rotate the pinion), a lock must be provided to hold the rack or the shaft while it is being clamped.

1.14.3. Work-holding devices for N/C Machines: Standard milling machine vises can be used to conveniently hold many workpieces on N/C machines. To ensure that each workpiece is located exactly in the same position, an adjustable locator is used. The adjustable locator will take care of any variation in sizes of workpieces. This arrangement is suitable only for small components. for a larger range of workpiece sizes, adjustable vises are available, as standard items. Many jobs can be held in 3- and 4-jaw chucks mounted on flat plate.

Base plate with grid pattern of holes: A base plate (square on rectangular) with a grid pattern of holes, Fig. 1.83, is especially suited as a work-holding device for N/C machine tools. The holes are usually on 25 mm (1") centres. Every other hole is tapped to hold down studs or screws. The remaining holes are jig-bored to 0.005 mm accuracy, and these holes hold dowel pins. The dowel pins act as locators and the studs and screws are used for clamping the work-piece on the plate. The grid plate, in turn, is bolted directly on the machine table. The grid plate system can be used for an infinite varieties of workpiece.

The bottom-left most hole is taken as the zero reference point, Fig. 1.83. With this, the grid pattern and hence the work-piece will be placed in the first-quadrant (upper right-hand quadrant). All the dimensions measured from the reference point will be positive in magnitude. The N/C machine tool is set-up by bring the machine spindle directly over the reference hole with the help of a dial indicator mounted on the spindle. The workpiece is then located and held on the grid plate in reference to the set up point.

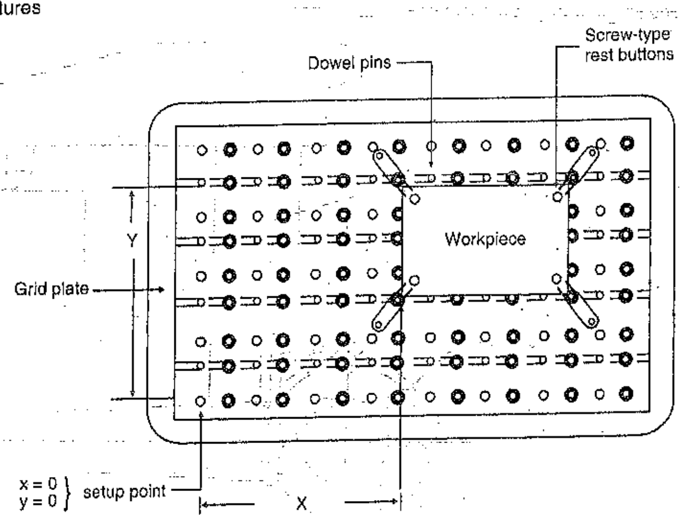


Fig. 1.83. A base plate with grid pattern of holes

One major drawback of grid plate is that the chips get accumulated in the holes. However, this can be avoided by keeping all the unused holes plugged.

Modular fixtures: These fixtures are commonly used on N/C machining centres. The fixtures comprise of modular components based on the grid principle (A module is an inter-changeable "plug in" item that may be combined with other inter-changeable items to form a complete unit, *i.e.*, a modular fixture is built up on the building block principle). The workpiece locators are located approximately on the grid pattern and then the workpiece is precision-machined under tape control. Dowel pins are not used. Fig. 1.84, shows a typical modular fixture.

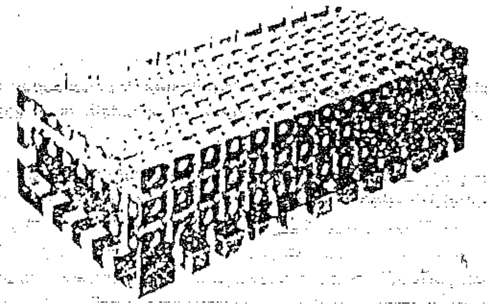


Fig. 1.84. Modular Fixture

1.15 Fundamentals of Jig and Fixture Design:

For designing a jig or a fixture the following data should be available to the designer:

1. Working drawing of the finished part, the specifications for its inspection and the working drawing of the blank. The term blank, here, refers to the starting work piece which will be machined to get the finished part. The blank may be in the form of: a casting, a stamping, a forging, a weldment, a blank cut from a rolled stock or a blank produced by Powder Metallurgy.

2. Operational sketches of the part for the preceding and given operations (if the jig or fixture is being designed for some intermediate operation).

3. A process planning sheet for the given part, which indicates the sequence and description of the operations to be performed on the blank, adopted datum features, machine tool and cutting tools to be used, cutting speeds, feeds and depths of cut, and the designed standard time per piece with the handling time for loading, clamping and unloading the part shown separately.

4. Standard for the components and units of machine tool jigs and fixtures, as well as a manual of standard designs of jigs and fixtures.

The part to be machined is located and clamped in a jig or a fixture. The datum features (mentioned above) refer to the totality of surfaces, lines and points of a part that are in contact with the location and clamping elements of a jig or a fixture, for machining of its various surfaces in a given operation. Here, the datum features will be called as "locating features or surfaces". The correct choice of locating surfaces is essential in planning a manufacturing process, because, wrong locating surfaces will result in great scrap losses.

Jig and fixture design should be tied up with process planning activities since the latter involve the selection of the datum features and establishing the sequence of machining operations with operational dimensions and tolerances for each operation. Also at this stage, the process engineer makes sketches that indicate how the part is to be located and clamped, selects the cutting variables, determines the standard time per piece from its elements, selects the cutting tools and the machine tools to be used.

5. In addition to the above information, it is necessary to know the basic dimensions of the machine tool with respect to the installation of the jig or fixture, that is, dimensions of the table, size and arrangement of the *T*-slots, minimum distance from the table to the spindle, size of spindle taper socket, etc., as well as the general condition of the machine tool.

The design of the jig or fixture is selected in accordance with the available production facilities and the required volume of production. Also, it is to be decided at this stage, whether the parts subjected to most wear in the jig or fixture are to be of renewable design.

With the above information available, the design of a jig or a fixture may follow the following steps:

(*i*) After deciding about the locating surfaces, and knowing about their accuracy and finish, the designer selects the most suitable type, size, number and general arrangement of the locating elements.

(*ii*) Next, the designer determines the cutting forces from the cutting variables (cutting speed and feed and depth of cut). Then the points of application and the required magnitudes of the clamping forces are determined. On the basis of planned time for clamping and releasing the part, the type of jig or fixture (Single-or multiple-piece), the shape and accuracy of the part, and the magnitude of the clamping forces, the type of clamping device is selected and its dimensions determined. At the same time, the type and size of the guiding elements are determined, as well as of the elements for checking the position of the cutting tool.

(*iii*) After that, the necessary auxiliary devices are selected, and their size and design are established on the basis of the weight and type of the part and the required machining accuracy. The proper standards are used for this purpose.

(*iv*) Next, to draw the general arrangement of the jig or the fixture a number of views of the part are drawn. The number of views will depend upon the complexity of the part. The elements of the jig or fixture are next consecutively drawn, where required, around the part. First, the locating elements are drawn then the clamping devices, tool guiding elements and the auxiliary devices. Finally the body is designed around the part as the element that links the other elements into an integral unit. The assembly drawing usually carries the overall dimensions and the dimensions that should be held to in assembling the jig or fixture and stipulates the assembly specifications. In making the assembly drawing and the part drawings, tolerances are established on the dimensions.

1.15.1. Standard Jig and Fixture Parts. We all know that the basic purpose of standardization of any manufactured items is to ensure interchangeability (which is the basis of mass production) and facilitate the manufacture of parts. It also facilitates maintenance and repair. Also, any item, mass produced, is much cheaper than the same item made as a single unit. A number of standardized component parts for the construction of jigs and fixtures are available commercially. Thus, when designing and manufacturing jigs and fixtures, maximum use should be made of standard, readily purchasable (or available in store) component parts. This will reduce the cost of design and manufacturing considerably. The designer should avoid, as far as possible, specifications of odd shaped or sized items that must be specially made.

The standardized component parts for jigs and fixtures include : drill bushings, washers, handles, various types of screws, studs and nuts, various types of knobs, jig straps, rest buttons or jig feet locating pins, fixture or mill keys, *T*-bolt and *T*-nut, Heel rest, brackets, toggles, came and wedges, etc. Many of these have already been sketched and discussed.

1.16. JIG AND FIXTURE CONSTRUCTION

The body of a jig or a fixture can be constructed by any of the following methods:

(1) Casting (2) Fabrication (3) Welding.

In the last two methods, the body is made up from separate parts. The advantages of the cast construction are :

(*i*) Jigs and fixtures with complicated shapes can be easily cast.

(*ii*) Cast iron has the property of absorbing and damping out the vibrations.

(*iii*) Any number of castings with the same characteristics can be made from one pattern.

(*iv*) If a cast jig or fixture drops down, it will probably break. It is not likely to bend and get out of alignment so as to result in the production of defective pieces.

Fabrication has the following advantages:

(*i*) Standard parts can be used to build up the body.

(*ii*) The jig or fixture can be built up quickly by using standard parts.

(*iii*) After use, the jig or fixture can be disassembled.

Welding has the following advantages:

(*i*) The jig or fixture can be constructed speedily.

(*ii*) The welded construction is cheaper.

(*iii*) Less machining is needed than for fabricated parts.

Comparison. Casting is the most popular method. It is costly both in time and money but it has the property of absorbing and damping out vibrations, a very useful characteristic for milling fixtures. Castings are usually heavy and where lightness is needed, welded or fabricated jig or fixture has the advantages. If a fabricated or welded jig or fixture gets dropped, it will get distorted. The defect will remain unnoticed until the device is used again when it produces defective piece. A cast construction will break rather than bend. For ease and quickness of manufacture, the order is : (*a*) Welded (*b*) Fabricated (*c*) Cast.

The welded construction must be "stress relieved" by heat treatment, to relieve all internal stresses induced during welding. Otherwise, the stresses will gradually distort the jig or fixture, as they relieve themselves. Only after stress relief, can finish machining be done on the construction, if accuracy is to be held. And, since machining must be done after assembly, it is sometimes difficult to machine an internal surface of a welded construction.

In a fabricated construction, all the component parts can be completely machined before assembly.

This is often an important enough advantage to offset the lesser cost of the welded construction. Often a combination of the two methods is successfully used. In the fabrication method, the construction is held together by screws and dowels. The screws serve only to hold the parts together with the dowels assuring accurate alignment. Two dowels are sufficient in any case, whereas the screws can be one or more as needed. A fabricated construction is easy to repair, because disassembly is comparatively easy. To facilitate disassembly and assembly, each dowel should be a press fit in one part and slip fit in the other.

Casting construction is sometimes used for the main body (especially for a jig) and other pieces are welded or assembled to it with screws and dowels.

1.17. MATERIALS FOR JIGS AND FIXTURES

S. No.	Description of the item	Recommended Material	Heat Treatment
.	Jig bush	40Cr 1	Oil hardened and tempered 40–45 HRC
		T90	–do– 50–58 HRC
		C14	Case hardened and tempered to 55–56 HRC
		15 Cr 65	–do–
		T21 5Cr 12	Oil hardened and tempered 58–63 HRC
2.	Washer	St 42	
		C 45	
		55 Cr 70	Oil hardened and tempered to 40–45 HRC
3.	Palmgrip	C 20	
		C 45	
		40Cr 1	Oil hardened and tempered 40 to 50 HRC
		Cast Iron grade 20	
		Aluminium alloy type C	
4.	Taper handle	C45	
5.	Ball handle	C45	
		Aluminium casting	
		Plastics	
6.	Tenon	C45	Oil hardened and tempered 36–45 HRC
7.	Set collars	C45	–do–
8.	Handle wheels	Gray cast Iron Grade 20	
		Aluminium Alloy Casting, A-24 M	
		Phenol Plastics	
9.	Strap clamp	C45	
		40 Cr	Oil hardened and tempered 40 – 50 HRC
		C14	Case hardened and temepered 52–62 HRC
10.	'C' washers	15 NiCr 1 MO 12	–do–
11.	Jig Button	T 90	Oil hardened and tempered 54–58 HRC
12.	Locking dig (Catle)	C 14	Case hardened and tempered 56–60 HRC
13.	Jig Feet	T 90	Oil hardened and tempered 54–58 HRC
14.	Welded V-block	St 42	Stress relieving after welding
		C45	–do–

15.	Base Plate	St 42	–do–
		C45	–do–
		Cast Iron grade 20	
16.	Locating Pin	C45	Oil hardened and tempered 55–61 HRC
		T 90	Case hardened and tempered 56–60 HRC
		15 Cr65	–do–
17.	Locating plugs	C14	–do–
		T215Cr 12	Oil hardened and tempered 58–62 HRC
18.	Swing and Table clamps	40 Cr 1	Oil hardened and tempered 40–45 HRC
19.	Clamp plate	40 Cr 1	–do–
20.	Screw/wing, Knurled;	C–45	
	Grub or socket head cap	St 42	
		40 Cr 1	Oil hardened and tempered 40–45 HRC
		Brass	
		Bronze	
		Aluminium Alloy	
21.	Thrust pad	St 42	
		40 Cr 1	Oil hardened and tempered 55–60 HRC
22.	V Block	Cast iron grade 20	
		C45	Surface flame hardened 40–45 HRC
		40 Cr 1	–do–
		C 14	Case hardened and tempered 56–60 HRC
23.	T Nut	St 42	
		C45	
		40 Cr 1	Oil hardened and tempered 42–45 HRC
24.	Plate, base, Housing	St 42	Stress relieving after welding if any
		C45	–do–
		Cast iron Grade 20	
25.	Jaws for vice	T 90	Oil hardened and tempered 52–58 HRC
26.	Claming plates and levers	C45	
		40 Cr 1	
27.	Key	C45	
		St 42	
28.	Hinge pin drilled	T 90	
29.	Foot (positioned) Plate	T 90	Oil hardened and tempered 54–58 HRC
30.	Foot Round	T 90	–do–

Note: Cast Iron, because of its damping capacity, is a very important material for the manufacture of jigs and fixtures. For heavy bodies, it is always economical to use Cast iron. Again complicated and detailed machining can be avoided. Where lightness is the main consideration, the materials can be alloys of aluminium and magnesium. In the case of screw operated feeding and clamping devices, phosphor bronze replacable nuts are used, screws being costly items. Nuts will wear out early and can be replaced. They reduce the wear of the screws and also provide corrosion resistance.